# Chas Campbell Gravitational Engine

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**Résumé**

Fundamental theoretical research on the Chas Campbell Gravitational Engine.

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# 1 Introduction

The Chas Campbell Gravitational Engine is considered in this work as a mechanical system whose operation involves gravitational forces, rotating masses, mechanical constraints, and electromechanical components.

The purpose of the present research is to establish a rigorous theoretical framework for analyzing the proposed mechanism. The analysis does not assume in advance that the system can produce a net excess of energy. Instead, the objective is to determine whether the proposed arrangement of masses, rotating elements, and energy-conversion components is consistent with the laws of classical mechanics and energy conservation.

The research therefore proceeds from explicit physical assumptions to a mathematical description of the system. The resulting model can subsequently be evaluated analytically and numerically under a range of operating conditions.

## 1.1 Research objective

The primary objective is to develop a mathematical and physical model of the Chas Campbell Gravitational Engine that is sufficiently explicit to permit independent analysis and numerical validation.

The investigation focuses on the following questions :

— How should the mechanical configuration of the proposed engine be represented mathematically ?
— What gravitational forces and torques act on the moving components ?
— How does the distribution of mass affect the rotational dynamics ?
— How is gravitational potential energy converted into mechanical kinetic energy during the motion of the system ?
— What role does the flywheel or rotating inertia play in storing and releasing mechanical energy ?
— How should the motor and generator components be represented in the complete electromechanical model ?
— What are the mechanical and electrical losses associated with a realistic implementation ?
— Does the complete system exhibit a net energy balance compatible with sustained cyclic operation ?

These questions define the principal scope of the theoretical investigation.

## 1.2 Research methodology

The analysis is organized as a sequence of increasingly detailed models.

First, the physical assumptions and idealizations are defined. This provides a common basis for the mathematical treatment and identifies the limitations of the model.

Second, the relevant mechanical variables are introduced. These include mass, position, height, angular position, angular velocity, moment of inertia, force, torque, and power.

Third, the equations governing gravitational and rotational dynamics are formulated. Gravitational potential energy and rotational kinetic energy are treated explicitly so that energy transfers within the system can be identified.

Fourth, the flywheel and electromechanical conversion stages are introduced. The purpose is to determine how energy is temporarily stored, transferred, and dissipated throughout the complete machine.

Finally, the resulting equations are implemented in numerical simulations. The simulations are intended to test the theoretical model over specified initial conditions and parameter ranges rather than to replace the physical analysis.

## 1.3 Physical scope

The initial analysis is based on classical mechanics and assumes a non-relativistic system operating in the terrestrial gravitational field.

The gravitational field is initially approximated as uniform over the dimensions of the machine. Under this approximation, the gravitational potential energy of a mass $m$ at height $h$ is written as

$$U_g = mgh, \tag{1}$$

where $g$ denotes the local gravitational acceleration.

For rotating bodies, the rotational kinetic energy is represented by

$$T_{\text{rot}} = \frac{1}{2} I \omega^2, \tag{2}$$

where $I$ is the moment of inertia about the relevant axis and $\omega$ is the angular velocity.

These expressions provide the fundamental energy terms used in the subsequent analysis.

## 1.4 Energy conservation

A central principle of this research is that gravitational potential energy must be treated as part of the energy balance of the complete system.

A decrease in gravitational potential energy can produce a corresponding increase in kinetic or mechanical energy. However, this conversion does not by itself imply the creation of additional energy.

For a complete cycle, the relevant quantities must therefore be evaluated over the entire sequence of mechanical states. In particular, the analysis must determine whether the system returns to an equivalent configuration after delivering useful work.

A temporary increase in rotational speed, torque, or output power is not sufficient to establish sustained net energy production. The complete energy balance must include the energy required to restore the system to its initial state as well as all relevant dissipative losses.

Consequently, the theoretical model is formulated so that the following quantity can ultimately be evaluated :

$$\Delta E_{\text{total}} = E_{\text{input}} - E_{\text{output}} - E_{\text{loss}}, \tag{3}$$

with the precise definition of each term depending on the system boundary selected for the analysis.

## 1.5 System boundaries

The definition of the system boundary is particularly important for the analysis of a gravitational machine.

At the mechanical level, the system may include the moving masses, supporting structures, shafts, gears, bearings, flywheels, and other mechanical components.

At the electromechanical level, the system may additionally include the motor, generator, electrical connections, power electronics, and associated loads.

The energy balance must be evaluated consistently for the selected system boundary. Energy transferred across that boundary must be identified as input or output rather than being treated as internally generated energy.

The distinction between internal energy transfer and external energy input is therefore maintained throughout the subsequent mathematical development.

### 1.6 Model hierarchy

The research uses a hierarchy of models rather than attempting to represent all physical phenomena simultaneously.

The first model is an ideal mechanical model intended to isolate the fundamental gravitational and rotational dynamics.

The second model introduces additional mechanical effects such as friction, aerodynamic drag, and transmission losses.

The third model incorporates the motor and generator and establishes the electromechanical energy flows.

The final model is intended to combine the relevant mechanical and electrical components into a complete machine representation suitable for numerical simulation.

This hierarchical approach makes it possible to determine which physical effects are responsible for changes in the predicted behavior of the system.

### 1.7 Validation strategy

The theoretical model must be validated through independent checks of its equations, dimensions, limiting cases, and numerical implementation.

Dimensional consistency is required for all equations. For example, the gravitational potential energy must have dimensions of energy, while torque must have dimensions of force multiplied by distance.

The model should also reproduce established physical behavior in limiting cases. For example, when gravitational coupling or external torque is removed, the corresponding equations should reduce to the expected conservation or free-motion relations.

Numerical simulations will subsequently be compared with analytical predictions wherever closed-form solutions are available.

The validation process is therefore intended to distinguish mathematical properties of the model from numerical artifacts.

### 1.8 Organization of the research

The remainder of this document is organized as follows.

Section 2 introduces the theoretical foundations required for the analysis, including the relevant principles of classical mechanics, gravitational potential energy, rotational dynamics, and energy conservation.

Section 3 defines the assumptions and limitations of the physical model.

Section 4 develops the mathematical model and introduces the principal variables and equations.

Section 5 analyzes the gravitational dynamics of the proposed mechanism.

Section 6 develops the rotational dynamics and associated torque equations.

Section 7 examines the flywheel and its role as a mechanical energy-storage element.

Section 8 introduces the motor and generator model and the corresponding electromechanical energy transfers.

Section 9 combines the individual components into a machine-level model.

Section 10 describes the numerical simulation methodology.

Section 11 presents the results obtained from the simulations.

Section 12 discusses the physical interpretation and limitations of those results.

Finally, Section 13 summarizes the conclusions of the theoretical investigation and identifies possible directions for further research.

# 2 Theoretical Foundations

The analysis of the Chas Campbell Gravitational Engine is based on the principles of classical mechanics, gravitational potential energy, rotational dynamics, and conservation of energy.

The purpose of this section is to establish the physical and mathematical principles used throughout the remainder of the research. The specific geometry and mechanical configuration of the proposed machine are introduced later, after the fundamental quantities and governing laws have been defined.

## 2.1 Classical mechanics

The mechanical behavior of the system is described within the framework of Newtonian mechanics.

For a particle of mass $m$ subjected to a resultant force $\mathbf{F}$, Newton's second law is

$$\mathbf{F} = m\mathbf{a}, \tag{4}$$

where $\mathbf{a}$ is the acceleration of the particle.

For a system containing several bodies, Newton's equations can be applied individually to each body or to the system as a whole, depending on the chosen system boundary.

The equations of motion provide the relationship between applied forces, mass, acceleration, and the resulting mechanical trajectory.

## 2.2 Position and velocity

The position of a particle or body is represented by the vector

$$\mathbf{r}(t). \tag{5}$$

Its velocity is defined as

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}, \tag{6}$$

and its acceleration is

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}. \tag{7}$$

These quantities describe translational motion and will be used when analyzing the trajectories of masses within the proposed mechanism.

## 2.3 Work and mechanical power

The infinitesimal mechanical work performed by a force $\mathbf{F}$ during an infinitesimal displacement $d\mathbf{r}$ is

$$dW = \mathbf{F} \cdot d\mathbf{r}. \tag{8}$$

The total work performed between two positions is therefore

$$W = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r}. \tag{9}$$

Mechanical power is the rate at which work is performed :

$$P = \frac{dW}{dt}. \tag{10}$$

For translational motion, this becomes

$$P = \mathbf{F} \cdot \mathbf{v}. \tag{11}$$

For rotational motion, the corresponding expression is

$$P = \tau\omega, \tag{12}$$

where $\tau$ is the applied torque and $\omega$ is the angular velocity.

These relationships are fundamental for determining the instantaneous power transferred between the different components of the machine.

### 2.4 Gravitational force

Near the surface of the Earth, the gravitational force acting on a mass $m$ is approximated by

$$\mathbf{F}_g = m\mathbf{g}, \tag{13}$$

where $\mathbf{g}$ is the local gravitational acceleration vector.

For the initial model, the gravitational acceleration is considered uniform and its magnitude is approximated by

$$g \simeq 9{,}81\,\mathrm{m/s^2}. \tag{14}$$

The corresponding force magnitude is therefore

$$F_g = mg. \tag{15}$$

The direction of the gravitational force is approximately vertical and downward.

This approximation is sufficient for a first-order analysis provided that the dimensions of the machine are small compared with the spatial scale over which the terrestrial gravitational field varies significantly.

### 2.5 Gravitational potential energy

The gravitational potential energy of a mass $m$ at height $h$ relative to a chosen reference level is

$$U_g = mgh. \tag{16}$$

For a system containing multiple masses, the total gravitational potential energy is

$$U_g = \sum_i m_i g h_i. \tag{17}$$

The change in gravitational potential energy between two configurations is

$$\Delta U_g = U_{g,2} - U_{g,1}. \tag{18}$$

Consequently,

$$\Delta U_g = \sum_i m_i g \left(h_{i,2} - h_{i,1}\right). \tag{19}$$

A reduction in gravitational potential energy can provide mechanical energy to the system. However, the corresponding energy must be accounted for in the complete energy balance.

In particular, gravitational potential energy does not constitute an independent source of energy. It represents energy associated with the configuration of the masses within the gravitational field.

## 2.6 Conservative forces

Gravity is treated as a conservative force within the terrestrial approximation used in this work.

For a conservative force, the work performed between two configurations depends only on the initial and final states and not on the path followed.

The relationship between gravitational force and potential energy can be written as

$$\mathbf{F}_g = -\nabla U_g. \tag{20}$$

In the one-dimensional vertical case,

$$F_g = -\frac{dU_g}{dh}. \tag{21}$$

Using

$$U_g = mgh, \tag{22}$$

one obtains

$$F_g = -mg, \tag{23}$$

when the positive vertical direction is defined upward.

This relationship provides the basis for deriving the gravitational forces and torques acting on the individual components of the proposed machine.

## 2.7 Kinetic energy

The translational kinetic energy of a body of mass $m$ moving with speed $v$ is

$$T_{\text{trans}} = \frac{1}{2}mv^2. \tag{24}$$

For a rigid body rotating about a fixed axis, the rotational kinetic energy is

$$T_{\text{rot}} = \frac{1}{2}I\omega^2, \tag{25}$$

where $I$ is the moment of inertia about the axis of rotation.

For a system containing both translational and rotational degrees of freedom, the total kinetic energy is obtained by summing the appropriate contributions.

## 2.8 Moment of inertia

The moment of inertia quantifies the resistance of a body to changes in its rotational motion.

For a discrete collection of point masses,

$$I = \sum_i m_i r_i^2, \tag{26}$$

where $r_i$ is the perpendicular distance of mass $m_i$ from the rotational axis.

For a continuous mass distribution,

$$I = \int r^2\, dm. \tag{27}$$

The moment of inertia depends not only on the total mass of the system but also on the spatial distribution of that mass.

This property is particularly important for the present research because changes in the position of masses relative to a rotational axis can modify the rotational response of the system.

## 2.9 Torque

The torque produced by a force $\mathbf{F}$ applied at position $\mathbf{r}$ relative to a chosen rotational axis is

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}. \tag{28}$$

Its magnitude is

$$\tau = rF \sin\theta, \tag{29}$$

where $\theta$ is the angle between the position vector and the applied force.
For a gravitational force acting on a mass,

$$\boldsymbol{\tau}_g = \mathbf{r} \times m\mathbf{g}. \tag{30}$$

The gravitational torque depends on both the magnitude of the gravitational force and the moment arm relative to the rotational axis.
The sign of the torque depends on the selected positive direction of rotation.

## 2.10 Rotational dynamics

For a rigid body rotating about a fixed axis with constant moment of inertia, the rotational equation of motion is

$$I\dot{\omega} = \sum \tau. \tag{31}$$

Since

$$\omega = \dot{\theta}, \tag{32}$$

the rotational equation may also be written as

$$I\ddot{\theta} = \sum \tau. \tag{33}$$

If the moment of inertia changes with time or configuration, the more general angular-momentum equation is

$$\frac{d}{dt}(I\omega) = \sum \tau. \tag{34}$$

This distinction will be important when analyzing configurations in which masses move relative to the rotational axis.

## 2.11 Angular momentum

The angular momentum of a rigid body rotating about a fixed axis is

$$L = I\omega. \tag{35}$$

The general equation governing angular momentum is

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}_{\text{ext}}, \tag{36}$$

where $\boldsymbol{\tau}_{\text{ext}}$ is the resultant external torque.
In the absence of external torque about the selected axis,

$$\frac{d\mathbf{L}}{dt} = 0, \tag{37}$$

and angular momentum is conserved.

This conservation law provides an additional independent method for checking the consistency of the rotational model.

## 2.12 Energy conservation

The total energy of an isolated physical system is conserved.

For the mechanical system considered in this research, the principal contributions are gravitational potential energy, translational kinetic energy, and rotational kinetic energy.

An idealized mechanical energy can therefore be represented as

$$E_{\text{mech}} = T_{\text{trans}} + T_{\text{rot}} + U_g. \tag{38}$$

For an ideal conservative system,

$$\frac{dE_{\text{mech}}}{dt} = 0. \tag{39}$$

Thus,

$$E_{\text{mech},1} = E_{\text{mech},2}. \tag{40}$$

When non-conservative forces or external energy transfers are present, the energy equation must be modified accordingly.

A general energy balance can be written as

$$E_{\text{final}} = E_{\text{initial}} + E_{\text{input}} - E_{\text{output}} - E_{\text{loss}}. \tag{41}$$

This equation will form one of the principal validation criteria for the complete machine model.

## 2.13 Energy over a complete cycle

The distinction between instantaneous power and net energy transfer over a complete cycle is essential.

The instantaneous mechanical power may be positive during one part of a cycle and negative during another part. Consequently, an apparent power output during part of the motion cannot by itself demonstrate sustained energy production.

For a periodic mechanical process, the net work over one complete cycle is

$$W_{\text{cycle}} = \oint \tau \, d\theta. \tag{42}$$

For a conservative system returning to its initial configuration, the net work associated with conservative internal forces over the complete cycle is zero :

$$\oint \tau_{\text{conservative}} \, d\theta = 0. \tag{43}$$

If dissipative effects are present, additional work must be supplied to compensate for the associated energy losses.

Therefore, the condition for sustained cyclic operation must be evaluated using the complete system energy balance rather than by examining a single portion of the trajectory.

### 2.14 Dissipative forces

Real mechanical systems contain dissipative mechanisms.

A simplified viscous rotational loss model can be represented by

$$\tau_{\text{damping}} = -c\omega, \tag{44}$$

where $c$ is an effective damping coefficient.

A simplified Coulomb friction model can be represented by

$$\tau_{\text{friction}} = -\tau_c \operatorname{sgn}(\omega), \tag{45}$$

where $\tau_c$ is an effective friction torque.

These models are approximations. More detailed representations may be introduced when experimental or manufacturer data become available.

Dissipative effects always need to be included when evaluating the performance of a physical implementation.

### 2.15 Flywheel energy storage

A flywheel stores mechanical energy in rotational motion.

For a flywheel with moment of inertia $I_f$ and angular velocity $\omega_f$, the stored kinetic energy is

$$E_f = \frac{1}{2} I_f \omega_f^2. \tag{46}$$

The change in stored energy between two rotational states is

$$\Delta E_f = \frac{1}{2} I_f \left(\omega_{f,2}^2 - \omega_{f,1}^2\right). \tag{47}$$

The flywheel can therefore absorb energy during one part of a cycle and release energy during another part.

However, this storage mechanism does not constitute energy generation. Over a complete cycle, the energy stored in the flywheel must be accounted for in the overall energy balance.

### 2.16 Electromechanical conversion

When an electric motor or generator is connected to the mechanical system, energy can be transferred between electrical and mechanical domains.

The mechanical power associated with a rotating shaft is

$$P_{\text{mech}} = \tau\omega. \tag{48}$$

For an ideal electromechanical converter,

$$P_{\text{electrical}} = P_{\text{mech}}. \tag{49}$$

For a real converter, losses result in

$$P_{\text{out}} = \eta P_{\text{in}}, \tag{50}$$

where

$$0 < \eta \leq 1. \tag{51}$$

The motor and generator are therefore treated as energy-conversion devices rather than independent energy sources.

Their detailed mathematical representation is developed later in the research.

### 2.17 Dimensional consistency

All equations used in the theoretical model must satisfy dimensional consistency.

For example, gravitational potential energy is

$$U_g = mgh, \tag{52}$$

with dimensions

$$[U_g] = \mathrm{kg\,m^2/s^2}. \tag{53}$$

This corresponds to the SI unit of joule.

Similarly, rotational kinetic energy is

$$T_{\mathrm{rot}} = \frac{1}{2} I\omega^2, \tag{54}$$

where

$$[I] = \mathrm{kg\,m^2} \tag{55}$$

and

$$[\omega] = \mathrm{rad/s}. \tag{56}$$

Since the radian is dimensionless in the SI system, the resulting dimension is again that of energy.

Dimensional analysis will be used throughout the subsequent mathematical development as a basic consistency check.

### 2.18 Limiting cases

A valid mathematical model should reproduce established physical behavior under appropriate limiting conditions.

Several limiting cases are particularly relevant :

- when gravitational effects are removed, the corresponding gravitational force and torque must vanish ;
- when mechanical losses are set to zero, the model must reduce to the ideal conservative case ;
- when the applied torque is zero, the rotational dynamics must be consistent with angular-momentum conservation ;
- when the flywheel inertia approaches zero, its contribution to stored rotational energy must also approach zero ;
- when the efficiency of an idealized converter approaches unity, conversion losses must approach zero.

These limiting cases provide useful tests for both analytical derivations and numerical implementations.

### 2.19 Implications for the present research

The theoretical principles established in this section impose important constraints on the interpretation of the proposed gravitational engine.

A change in gravitational potential energy can produce mechanical work, and a rotating mass can store substantial kinetic energy. Likewise, a flywheel can redistribute mechanical power over time, while a motor or generator can transfer energy between electrical and mechanical forms.

None of these mechanisms, considered individually, violates conservation of energy.

The central question of the subsequent analysis is therefore whether the specific geometry and operating cycle proposed for the Chas Campbell Gravitational Engine can produce a sustained mechanical output after all energy transfers required to restore the system to its initial state are included.

The answer must follow from the equations of motion, energy balances, and numerical or experimental evidence developed in the subsequent sections.

# 3 Assumptions

The theoretical model is developed under a set of explicit assumptions. These assumptions define the physical system to be studied, the mathematical idealizations adopted, and the limits within which the subsequent analysis is intended to apply.

The purpose of these assumptions is not to establish the validity of the Chas Campbell Gravitational Engine a priori. Rather, they provide a clearly defined framework from which the proposed mechanism can be analyzed using classical mechanics, gravitational potential energy, rotational dynamics, and energy conservation.

## 3.1 Mechanical assumptions

The machine is modeled as a mechanical system composed of rigid bodies, rotating elements, support structures, and a flywheel or equivalent rotating inertia.

Unless otherwise specified, the principal mechanical components are assumed to behave as rigid bodies. Deformation of structural components is neglected in the first-order model.

The following assumptions are adopted :

— mechanical components have well-defined masses and moments of inertia ;
— bearings and mechanical supports maintain the prescribed geometric constraints ;
— the relevant rotating elements share a common reference axis when represented by the reduced rotational model ;
— structural deformation is neglected unless explicitly introduced in a subsequent model ;
— backlash, clearances, and microscopic mechanical imperfections are initially neglected ;
— friction and other dissipative effects are introduced separately as loss mechanisms.

These assumptions permit the machine to be represented initially as a deterministic mechanical system with a finite number of generalized coordinates.

## 3.2 Gravitational assumptions

The gravitational field is initially assumed to be uniform over the spatial extent of the machine.

The gravitational acceleration is therefore represented by the constant

$$g \simeq 9{,}81\,\mathrm{m/s^2}. \tag{57}$$

Under this approximation, the gravitational potential energy of a mass $m$ located at height $h$ relative to a chosen reference level is

$$U_g = mgh. \tag{58}$$

For a system containing several masses, the total gravitational potential energy is written as

$$U_g = \sum_i m_i g h_i. \tag{59}$$

The choice of the zero level for gravitational potential energy is arbitrary ; only differences in potential energy are physically relevant.

The uniform-field approximation is appropriate provided that the dimensions of the machine remain sufficiently small compared with the characteristic length scale over which the terrestrial gravitational field changes appreciably.

### 3.3 Rotational assumptions

Rotational motion is described using classical rigid-body mechanics.

For a rotating body with moment of inertia $I$ and angular velocity $\omega$, the rotational kinetic energy is

$$T_{\mathrm{rot}} = \frac{1}{2} I \omega^2. \tag{60}$$

The corresponding angular momentum about the rotational axis is

$$L = I\omega. \tag{61}$$

When the moment of inertia is constant, the rotational equation of motion takes the form

$$I\dot{\omega} = \sum \tau, \tag{62}$$

where $\tau$ denotes the applied torque about the relevant axis.

For a system in which the moment of inertia varies with time or configuration, the more general expression

$$\frac{d}{dt}(I\omega) = \sum \tau \tag{63}$$

is used.

The distinction between these two cases is important because changes in mass distribution may modify the rotational dynamics without necessarily representing the creation of mechanical energy.

### 3.4 Energy assumptions

The analysis is based on the conservation of energy.

For an isolated mechanical system, the total mechanical energy is written as

$$E_{\mathrm{mech}} = T_{\mathrm{trans}} + T_{\mathrm{rot}} + U_g + U_{\mathrm{other}}, \tag{64}$$

where the individual terms represent translational kinetic energy, rotational kinetic energy, gravitational potential energy, and any other conservative potential-energy contributions included in the model.

For a real machine, dissipative processes must also be considered. The available mechanical energy may therefore be represented schematically as

$$E_{\mathrm{out}} = E_{\mathrm{in}} + \Delta U_g + \Delta T - E_{\mathrm{loss}}, \tag{65}$$

where $E_{\mathrm{loss}}$ represents energy dissipated through friction, electrical resistance, aerodynamic drag, deformation, vibration, heat, and other irreversible processes.

No assumption of net energy creation is made in the present model.

In particular, gravitational potential energy is treated as an energy-storage term rather than as an independent energy source. Any sustained cyclic operation must therefore be evaluated by comparing the energy supplied to and extracted from the complete system over a full operating cycle.

## 3.5 Motor and generator assumptions

Where an electric motor or generator is included in the model, the electromechanical conversion process is represented explicitly.

For an ideal rotational electromechanical converter, mechanical power is

$$P_{\text{mech}} = \tau\omega. \tag{66}$$

For a real converter, the electrical and mechanical powers are related through an efficiency parameter $\eta$ satisfying

$$0 < \eta \leq 1. \tag{67}$$

The idealized conversion relations will initially be used to establish the fundamental dynamics. Losses and non-ideal conversion efficiencies will then be introduced in the corresponding machine model.

The motor and generator are not assumed to provide a net source of energy. They are treated as devices that convert energy between electrical and mechanical forms.

## 3.6 Flywheel assumptions

The flywheel is modeled initially as a rigid rotating body characterized by its mass distribution and moment of inertia.

Its rotational kinetic energy is

$$E_{\text{flywheel}} = \frac{1}{2} I_f \omega_f^2, \tag{68}$$

where $I_f$ is the flywheel moment of inertia and $\omega_f$ is its angular velocity.

For a flywheel whose angular velocity changes between $\omega_1$ and $\omega_2$, the corresponding change in stored rotational energy is

$$\Delta E_{\text{flywheel}} = \frac{1}{2} I_f \left(\omega_2^2 - \omega_1^2\right). \tag{69}$$

The flywheel is therefore considered an energy-storage element. Its ability to smooth power fluctuations or temporarily release stored energy must not be interpreted as generation of net energy.

## 3.7 Loss assumptions

The first-order analytical model may initially neglect secondary losses in order to isolate the fundamental mechanism.

However, the complete physical model is expected to include at least the following loss mechanisms :

— bearing and mechanical friction ;
— aerodynamic drag ;
— electrical resistance ;
— motor and generator conversion losses ;
— structural deformation and vibration ;
— losses associated with mechanical transmission.

The total dissipated power can subsequently be represented by

$$P_{\text{loss}} = P_{\text{friction}} + P_{\text{aero}} + P_{\text{electrical}} + P_{\text{conversion}} + P_{\text{other}}. \tag{70}$$

The ideal model and the loss-inclusive model will be treated separately so that the effect of each loss mechanism can be identified.

### 3.8 Initial and boundary conditions

The state of the machine is defined by the positions, velocities, angular positions, angular velocities, and relevant electrical variables required by the selected mathematical model.

Initial conditions must be specified before any numerical simulation is performed.

For a generic rotating component, the initial state may be represented by

$$\theta(0) = \theta_0, \qquad \omega(0) = \omega_0. \tag{71}$$

Additional initial conditions will be introduced when multiple degrees of freedom or electrical states are included.

The boundary conditions must also specify the mechanical constraints imposed by the support structure and the interaction between the machine and its environment.

### 3.9 Model limitations

The assumptions introduced above define an idealized theoretical model and therefore impose limitations on the conclusions that can be drawn from it.

In particular :

— the uniform gravitational-field approximation does not describe variations in gravitational acceleration over large distances ;
— rigid-body modeling does not capture structural deformation ;
— idealized bearings do not reproduce all real mechanical losses ;
— simplified motor and generator models do not capture every electromechanical effect ;
— numerical simulations are dependent on the accuracy of the parameters and initial conditions supplied to them.

Most importantly, the mathematical model must distinguish between transient energy release and sustained cyclic energy production. A temporary increase in output power or rotational speed is not, by itself, evidence of a net energy gain.

The subsequent chapters therefore evaluate the proposed mechanism through explicit equations of motion and energy balances rather than assuming the existence of a net energy surplus.

## 4 Mathematical Model

This chapter defines the mathematical framework used to represent the Chas Campbell Gravitational Engine as a coupled mechanical and electromechanical system. The purpose of the model is to establish explicit state variables, governing equations, constraints, and energy-flow relationships that can subsequently be implemented in numerical simulations.

The mathematical model is formulated without assuming that the proposed machine produces net positive energy. Such a conclusion must follow, if at all, from the complete dynamical and energy analysis.

### 4.1 System Representation

The machine is represented as a collection of mechanically coupled subsystems. Depending on the level of abstraction, these subsystems may include :

— gravitationally loaded masses or moving elements ;
— rotating elements ;
— one or more flywheels ;
— shafts and mechanical couplings ;
— an electric motor ;
— an electric generator ;
— mechanical and electrical loads.

The generalized state of the system may be written as

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{q}(t) \\ \dot{\mathbf{q}}(t) \\ \boldsymbol{\omega}(t) \\ \mathbf{i}(t) \end{bmatrix}, \tag{72}$$

where $\mathbf{q}$ denotes the generalized positional coordinates, $\dot{\mathbf{q}}$ their corresponding velocities, $\boldsymbol{\omega}$ the angular velocities of the rotating elements, and $\mathbf{i}$ the electrical current variables.

The exact dimension of $\mathbf{x}$ depends on the mechanical configuration selected for the final machine model.

## 4.2 Generalized Coordinates

For each independently moving mechanical degree of freedom, a generalized coordinate is introduced. A translational coordinate is denoted by

$$x_i(t), \tag{73}$$

while a rotational coordinate is denoted by

$$\theta_j(t). \tag{74}$$

The corresponding translational and angular velocities are

$$v_i(t) = \dot{x}_i(t) \tag{75}$$

and

$$\omega_j(t) = \dot{\theta}_j(t). \tag{76}$$

The corresponding accelerations are

$$a_i(t) = \ddot{x}_i(t) \tag{77}$$

and

$$\alpha_j(t) = \ddot{\theta}_j(t). \tag{78}$$

These variables provide the basis for deriving the equations of motion.

## 4.3 Gravitational Potential Energy

For a mass $m_i$ located at height $h_i$ in a uniform gravitational field, the gravitational potential energy is

$$U_{g,i} = m_i g h_i, \tag{79}$$

where $g$ is the local gravitational acceleration.

For a system containing several masses, the total gravitational potential energy is

$$U_g = \sum_i m_i g h_i. \tag{80}$$

If the height of a mass is a function of a generalized coordinate, then

$$h_i = h_i(\mathbf{q}), \tag{81}$$

and consequently

$$U_g = \sum_i m_i g h_i(\mathbf{q}). \tag{82}$$

The generalized gravitational force associated with coordinate $q_j$ is therefore obtained from

$$Q_{g,j} = -\frac{\partial U_g}{\partial q_j}. \tag{83}$$

This formulation ensures that gravitational forces are derived consistently from the potential-energy description.

### 4.4 Translational Dynamics

For a translational mass $m_i$, Newton's second law gives

$$m_i \ddot{x}_i = \sum F_i. \tag{84}$$

A simplified vertical equation for a mass subject to gravity and an additional applied force may be written as

$$m_i \ddot{h}_i = F_i - m_i g - F_{\text{loss},i}, \tag{85}$$

where $F_i$ represents an applied force and $F_{\text{loss},i}$ represents resistive forces.

Possible loss mechanisms include friction, bearing resistance, aerodynamic drag, deformation, and other mechanical dissipation.

### 4.5 Rotational Dynamics

For a rotating body with moment of inertia $I_j$, the rotational equation of motion is

$$I_j \dot{\omega}_j = \sum \tau_j. \tag{86}$$

Equivalently,

$$I_j \ddot{\theta}_j = \sum \tau_j. \tag{87}$$

The net torque may be decomposed into driving, resisting, gravitational, motor, generator, and coupling contributions :

$$I_j \dot{\omega}_j = \tau_{\text{drive}} + \tau_{\text{grav}} + \tau_{\text{motor}} - \tau_{\text{generator}} - \tau_{\text{loss}}. \tag{88}$$

The signs of individual terms depend on the chosen coordinate convention.

### 4.6 Torque–Force Relationship

When a tangential force $F$ acts at a radius $r$, the corresponding torque is

$$\tau = rF. \tag{89}$$

For a variable effective radius, the relationship becomes

$$\tau(t) = r(t)F(t). \tag{90}$$

This relationship is important when the geometry of the proposed mechanism causes the effective lever arm to change during operation.

If the force itself depends on position, then the instantaneous torque can be expressed as

$$\tau(\theta) = r(\theta)F(\theta). \tag{91}$$

The mechanical work associated with a rotational displacement is

$$W = \int_{\theta_1}^{\theta_2} \tau(\theta)\, d\theta. \tag{92}$$

### 4.7 Rotational Kinetic Energy

The kinetic energy of a rotating rigid body is

$$T_{\text{rot}} = \frac{1}{2} I\omega^2. \tag{93}$$

For several rotating components,

$$T_{\text{rot}} = \sum_j \frac{1}{2} I_j \omega_j^2. \tag{94}$$

The total rotational kinetic energy therefore depends on both the moments of inertia and the angular velocities of the rotating components.

### 4.8 Translational Kinetic Energy

For a translating mass, the kinetic energy is

$$T_{\text{trans}} = \frac{1}{2} mv^2. \tag{95}$$

For several translating masses,

$$T_{\text{trans}} = \sum_i \frac{1}{2} m_i v_i^2. \tag{96}$$

The total mechanical kinetic energy can consequently be written as

$$T = T_{\text{trans}} + T_{\text{rot}}. \tag{97}$$

### 4.9 Total Mechanical Energy

The ideal mechanical energy of the system is

$$E_{\text{mech}} = T + U_g + U_{\text{other}}, \tag{98}$$

where $U_{\text{other}}$ represents additional potential-energy terms when required by the mechanical configuration.

For an ideal conservative mechanical system without external energy exchange,

$$\frac{dE_{\text{mech}}}{dt} = 0. \tag{99}$$

For the actual machine, dissipative effects and electromechanical energy transfers must be included.

A more general energy balance is therefore

$$\frac{dE_{\text{mech}}}{dt} = P_{\text{in}} - P_{\text{out}} - P_{\text{loss}}, \tag{100}$$

where $P_{\text{in}}$ is externally supplied power, $P_{\text{out}}$ is useful mechanical or electrical output power, and $P_{\text{loss}}$ represents dissipative losses.

### 4.10 Flywheel Representation

A flywheel is represented by its moment of inertia $I_f$ and angular velocity $\omega_f$.
Its stored rotational kinetic energy is

$$E_f = \frac{1}{2} I_f \omega_f^2. \tag{101}$$

The corresponding angular momentum is

$$L_f = I_f \omega_f. \tag{102}$$

For a constant moment of inertia,

$$\frac{dE_f}{dt} = I_f \omega_f \dot{\omega}_f. \tag{103}$$

Since mechanical power associated with a torque is

$$P_{\text{mech}} = \tau\omega, \tag{104}$$

the flywheel equation can also be written as

$$\frac{dE_f}{dt} = \tau_f \omega_f. \tag{105}$$

A flywheel therefore acts as an energy-storage element. It does not, by itself, constitute an independent source of energy.

### 4.11 Motor Model

For an idealized electric motor, the mechanical power delivered to the shaft is

$$P_{\text{motor}} = \tau_m \omega_m. \tag{106}$$

A simplified efficiency relationship is

$$P_{\text{mech,motor}} = \eta_m P_{\text{elec,in}}, \tag{107}$$

where $\eta_m$ is the motor efficiency.
Consequently,

$$\tau_m \omega_m = \eta_m V_m I_m \tag{108}$$

for a simplified steady-state representation.
A more detailed motor model may subsequently introduce winding resistance, back electromotive force, inductance, and electromagnetic torque.

### 4.12 Generator Model

The generator converts mechanical power into electrical power.
The mechanical input power is

$$P_{\text{gen,mech}} = \tau_g \omega_g. \tag{109}$$

For an efficiency $\eta_g$, the electrical output power is

$$P_{\text{elec,out}} = \eta_g P_{\text{gen,mech}}. \tag{110}$$

Thus,

$$P_{\text{elec,out}} = \eta_g \tau_g \omega_g. \tag{111}$$

The electromagnetic reaction torque exerted by the generator on the mechanical system must be included in the mechanical equation of motion.

## 4.13 Electromechanical Coupling

When a motor and generator are mechanically coupled through a common shaft, the shaft dynamics may be represented by

$$I_{\text{eq}}\dot{\omega} = \tau_m - \tau_g - \tau_{\text{loss}}, \tag{112}$$

where $I_{\text{eq}}$ is the equivalent rotational inertia.

If multiple rotating components are rigidly coupled, their equivalent moment of inertia referred to the same shaft can be approximated by

$$I_{\text{eq}} = \sum_j I_j \tag{113}$$

for direct coupling.

For systems involving gears or other transmission ratios, the individual inertias must instead be referred to a common shaft using the appropriate kinematic transformation.

## 4.14 Preliminary Analytical Model

The following equations constitute an initial analytical framework proposed for the investigation of the Chas Campbell Gravitational Engine. At this stage of the research, these equations are introduced as working hypotheses and exploratory relationships. They are retained in their initial form so that their physical validity, dimensional consistency, and applicability can be assessed in subsequent chapters.

### 4.14.1 Fundamental Principle of Rotating Dynamics

The tangential component of the resultant forces is written as

$$F_t = m a_t. \tag{114}$$

The moment of force with respect to the origin is written as

$$C = F_t r. \tag{115}$$

The tangential acceleration is written as

$$a_t = r\frac{\mathrm{d}\omega}{\mathrm{d}t}. \tag{116}$$

### 4.14.2 Gravitational Energy of a Simple Pendulum

The gravitational potential energy of the simple pendulum is written as

$$E_p = mgr\left(1 - \cos(\theta)\right). \tag{117}$$

### 4.14.3 Kinetic Energy of a Simple Pendulum

The kinetic energy is written as

$$E_c = \frac{1}{2} m V^2. \tag{118}$$

Using $V = \omega r$, this expression is written as

$$E_c = \frac{1}{2} m \left(\omega r\right)^2. \tag{119}$$

### 4.14.4 Energy Conservation of a Simple Pendulum

The mechanical energy of the system is written as

$$E_m = E_c + E_p. \tag{120}$$

Consequently,

$$E_m = \frac{1}{2} m \left(\omega r\right)^2 + mgr\left(1 - \cos(\theta)\right). \tag{121}$$

### 4.14.5 Engine Torque and Power

The motor power is written as

$$P_{motor} = C_{motor}\omega_{motor}. \tag{122}$$

### 4.14.6 Pulley–Belt Ratio

The pulley-belt or drive ratio is written as

$$\frac{d_{out}}{d_{in}} = \frac{\omega_{in}}{\omega_{out}} = \frac{C_{out}}{C_{in}}. \tag{123}$$

### 4.14.7 Preliminary Flywheel Energy Model

The kinetic energy of the flywheel is written as

$$E_{c,flywheel} = M_{flywheel}\left(\omega_{flywheel} r_{flywheel}\right)^2. \tag{124}$$

The potential energy of the flywheel is written as

$$E_{p,flywheel} = 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{125}$$

The mechanical energy of the flywheel is written as

$$E_{m,flywheel} = M_{flywheel}\left(\omega_{flywheel} r_{flywheel}\right)^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{126}$$

Using the time derivative of the angular position gives

$$E_{m,flywheel} = M_{flywheel}\left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} r_{flywheel}\right)^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{127}$$

Equivalently,

$$E_{m,flywheel} = M_{flywheel} \left( \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \right)^2 r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{128}$$

The exploratory model assumes

$$E_{m,flywheel} = \text{constant.} \tag{129}$$

Therefore,

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 0. \tag{130}$$

The initial derivative is written as

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 2M_{flywheel} \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel} \cos\left(\theta_{flywheel}(t)\right). \tag{131}$$

The expanded expression is written as

$$2M_{flywheel} \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel} \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \cos\left(\theta_{flywheel}(t)\right) = 0. \tag{132}$$

With

$$\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \neq 0, \tag{133}$$

the exploratory equation of motion is written as

$$\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} + \frac{g}{r_{flywheel}} \cos\left(\theta_{flywheel}(t)\right) = 0. \tag{134}$$

The characteristic angular speed is written as

$$\omega_{flywheel} = \sqrt{\frac{g}{r_{flywheel}}}. \tag{135}$$

#### 4.14.8 Preliminary Motor Parameters

The angular speed of the motor in revolutions per minute is introduced as

$$\omega_{motor} = 1500. \tag{136}$$

The motor torque is written as

$$C_{motor} = 2M_{flywheel} g r_{flywheel}. \tag{137}$$

The mechanical power is written as

$$P_{m,motor} = C_{motor} \omega_{motor}. \tag{138}$$

Consequently,

$$P_{m,motor} = 2M_{flywheel} g r_{flywheel} \omega_{motor}. \tag{139}$$

If there is no resistance, the electrical power is written as

$$P_{e,motor} = P_{m,motor}, \tag{140}$$

and therefore

$$P_{e,motor} = 2M_{flywheel} g r_{flywheel} \omega_{motor}. \tag{141}$$

Let

$$d_{pulley,motor} \tag{142}$$

denote the diameter of the motor pulley in metres.

#### 4.14.9 Preliminary Electrical Generator Parameters

The angular speed of the generator in revolutions per minute is written as

$$\omega_{generator} = 60 f_{generator}. \tag{143}$$

For $f_{generator} = 50$, this gives

$$\omega_{generator} = 60 \times 50 \tag{144}$$

and

$$\omega_{generator} = 3000. \tag{145}$$

The electrical power is written as

$$P_{e,generator} = E_{e,generator} \omega_{generator}. \tag{146}$$

Consequently,

$$E_{e,generator} = \frac{P_{e,generator}}{\omega_{generator}}. \tag{147}$$

The preliminary model further assumes

$$E_{e,generator} = E_{m,flywheel}. \tag{148}$$

Thus,

$$M_{flywheel} \left( \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \right)^2 r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right) = \frac{P_{e,generator}}{\omega_{generator}}. \tag{149}$$

Let

$$d_{pulley,generator} \tag{150}$$

denote the diameter of the generator pulley in metres.

#### 4.14.10 Preliminary Flywheel and Pulley Parameters

The angular speed of the flywheel in revolutions per minute is written as

$$\omega_{flywheel} = \frac{d_{pulley,motor}\omega_{motor}}{d_{pulley,flywheel}}. \tag{151}$$

The pulley diameter of the flywheel is written as

$$d_{pulley,flywheel} = \frac{\omega_{generator}d_{pulley,generator}}{\omega_{flywheel}}. \tag{152}$$

The generator pulley diameter is written as

$$d_{pulley,generator} = \frac{\omega_{flywheel}d_{pulley,flywheel}}{\omega_{generator}}. \tag{153}$$

The motor pulley diameter is written as

$$d_{pulley,motor} = \frac{\omega_{flywheel}d_{pulley,flywheel}}{\omega_{motor}}. \tag{154}$$

With

$$\theta_{flywheel}(t) = \frac{\pi}{2}, \tag{155}$$

the flywheel mass is written as

$$M_{flywheel} = \frac{P_{e,generator}}{2\omega_{generator}g r_{flywheel}}. \tag{156}$$

### 4.15 Dissipative Effects

A first-order representation of rotational friction may be expressed as

$$\tau_{\text{loss}} = b\omega, \tag{157}$$

where $b$ is a viscous friction coefficient.
A more complete model may include Coulomb friction :

$$\tau_{\text{loss}} = b\omega + \tau_c \,\text{sgn}(\omega), \tag{158}$$

where $\tau_c$ is the Coulomb-friction torque.
Aerodynamic losses may also be approximated by

$$\tau_{\text{aero}} = k_a\omega^2 \,\text{sgn}(\omega), \tag{159}$$

where $k_a$ is an aerodynamic-loss coefficient.
The appropriate loss model will be determined experimentally or from validated engineering estimates where experimental data are not yet available.

## 4.16 Lagrangian Formulation

For a generalized mechanical system, the equations of motion can be formulated using the Lagrangian

$$\mathcal{L} = T - U, \tag{160}$$

where $T$ is the total kinetic energy and $U$ is the total potential energy.

For each generalized coordinate $q_j$, the Euler–Lagrange equation is

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j}\right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_j, \tag{161}$$

where $Q_j$ represents non-conservative generalized forces.

This formulation provides a systematic method for deriving the equations of motion when the machine contains several coupled degrees of freedom.

## 4.17 State-Space Form

For numerical simulation, the mechanical and electrical equations can be expressed in state-space form :

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, \mathbf{p}, t), \tag{162}$$

with

$$\mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}, \mathbf{p}, t), \tag{163}$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the external input vector, $\mathbf{p}$ is the vector of physical parameters, and $\mathbf{y}$ is the vector of measured or calculated outputs.

A representative state vector is

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} & \dot{\mathbf{q}} & \boldsymbol{\omega} & \mathbf{i} \end{bmatrix}^T. \tag{164}$$

This formulation will allow the model to be integrated numerically in the simulation stage of the research.

## 4.18 Energy-Balance Constraint

A fundamental constraint of the model is conservation of energy.

For a complete closed system, the total energy balance can be written as

$$E_{\text{initial}} + E_{\text{external,in}} = E_{\text{final}} + E_{\text{external,out}} + E_{\text{dissipated}}. \tag{165}$$

For a cyclic operation in which the machine returns to the same mechanical state,

$$\Delta E_{\text{stored}} = 0. \tag{166}$$

Consequently, the net energy available for external output over a complete cycle must be accounted for by the external energy supplied to the system, less all losses.

This condition is essential for evaluating whether an apparent mechanical or electrical surplus represents a genuine net energy gain or merely a temporary release of previously stored energy.

### 4.19 Model Verification Requirements

Before the model can be used to support physical conclusions, the following quantities must be independently verified :

— masses and mass distributions ;
— moments of inertia ;
— geometric dimensions ;
— gravitational force contributions ;
— friction and mechanical losses ;
— motor electrical input ;
— generator electrical output ;
— shaft torque ;
— angular velocity ;
— transient and steady-state energy balances.

The mathematical model developed in this chapter therefore provides a framework rather than a presupposed result. Subsequent chapters will specialize these equations to the gravitational, rotational, flywheel, motor-generator, and complete machine subsystems.

# 5 Gravitational Dynamics

This chapter develops the preliminary gravitational model used in the investigation of the Chas Campbell Gravitational Engine.

The objective of this chapter is to describe how gravity may contribute to the motion of the mechanical system and how gravitational potential energy is exchanged with kinetic energy during operation.

At this stage of the research, several equations are intentionally retained in their original form as working hypotheses. Their physical validity, dimensional consistency, and applicability to the proposed machine will be examined in subsequent chapters through analytical derivations, numerical simulations, and complete energy-balance verification.

### 5.1 Uniform Gravitational Field

The Earth gravitational field is assumed to be uniform over the entire machine volume.

The gravitational acceleration is denoted by

$$g. \tag{167}$$

The corresponding gravitational force acting on a mass $m$ is

$$F_g = mg. \tag{168}$$

For multiple masses, the total gravitational force is

$$F_{g,total} = \sum_i m_i g. \tag{169}$$

The present analysis assumes that local variations of the Earth's gravitational field are negligible.

### 5.2 Simple Pendulum Representation

As a preliminary approximation, a gravitating moving element may be represented as a simple pendulum.

The angular position is denoted by

$$\theta(t). \tag{170}$$

The pendulum length is denoted by

$$r. \tag{171}$$

The mass is denoted by

$$m. \tag{172}$$

The angular velocity is

$$\omega(t) = \frac{\mathrm{d}\theta(t)}{\mathrm{d}t}. \tag{173}$$

The angular acceleration is

$$\alpha(t) = \frac{\mathrm{d}^2\theta(t)}{\mathrm{d}t^2}. \tag{174}$$

### 5.3 Fundamental Principle of Rotating Dynamics

The tangential component of the resultant force is written as

$$F_t = ma_t. \tag{175}$$

The moment of force with respect to the origin is written as

$$C = F_t r. \tag{176}$$

The tangential acceleration is written as

$$a_t = r\frac{\mathrm{d}\omega}{\mathrm{d}t}. \tag{177}$$

Combining the previous equations gives

$$C = mr\left(r\frac{\mathrm{d}\omega}{\mathrm{d}t}\right). \tag{178}$$

This relation provides a first-order rotational representation for the gravitationally driven motion.

### 5.4 Gravitational Potential Energy of a Simple Pendulum

The gravitational potential energy is written as

$$E_p = mgr\left(1 - \cos(\theta)\right). \tag{179}$$

This relationship is used as a preliminary representation of the gravitational energy stored by the pendulum.

The height variation associated with the pendulum motion is therefore represented through the angular coordinate $\theta$.

## 5.5 Kinetic Energy of a Simple Pendulum

The kinetic energy is written as

$$E_c = \frac{1}{2} m V^2. \tag{180}$$

The tangential velocity can be written as

$$V = \omega r. \tag{181}$$

Consequently,

$$E_c = \frac{1}{2} m \left(\omega r\right)^2. \tag{182}$$

## 5.6 Mechanical Energy of a Simple Pendulum

The total mechanical energy of the system is written as

$$E_m = E_c + E_p. \tag{183}$$

Substituting the previous equations gives

$$E_m = \frac{1}{2} m \left(\omega r\right)^2 + mgr\left(1 - \cos(\theta)\right). \tag{184}$$

In the preliminary model, the mechanical energy is considered a useful quantity for tracking the exchanges between gravitational energy and kinetic energy.

## 5.7 Preliminary Flywheel Gravitational Model

The flywheel is represented by

$$M_{flywheel}, \tag{185}$$

$$r_{flywheel}, \tag{186}$$

and

$$\theta_{flywheel}(t). \tag{187}$$

The angular velocity is

$$\omega_{flywheel} = \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}. \tag{188}$$

## 5.8 Flywheel Kinetic Energy

The kinetic energy of the flywheel is written as

$$E_{c,flywheel} = M_{flywheel}\left(\omega_{flywheel} r_{flywheel}\right)^2. \tag{189}$$

## 5.9 Flywheel Potential Energy

The gravitational potential energy is written as

$$E_{p,flywheel} = 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{190}$$

### 5.10 Flywheel Mechanical Energy

The mechanical energy of the flywheel is written as

$$E_{m,flywheel} = M_{flywheel}\left(\omega_{flywheel} r_{flywheel}\right)^2 + 2M_{flywheel} g r_{flywheel}\sin\left(\theta_{flywheel}(t)\right). \quad (191)$$

Using

$$\omega_{flywheel} = \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}, \quad (192)$$

the previous equation becomes

$$E_{m,flywheel} = M_{flywheel}\left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} r_{flywheel}\right)^2 + 2M_{flywheel} g r_{flywheel}\sin\left(\theta_{flywheel}(t)\right). \quad (193)$$

or

$$E_{m,flywheel} = M_{flywheel}\left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\right)^2 r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel}\sin\left(\theta_{flywheel}(t)\right). \quad (194)$$

### 5.11 Preliminary Conservation Relation

The preliminary analytical model assumes

$$E_{m,flywheel} = constant. \quad (195)$$

Consequently,

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 0. \quad (196)$$

Differentiating the mechanical energy gives

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 2M_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel}\cos\left(\theta_{flywheel}(t)\right). \quad (197)$$

The expanded form is written as

$$2M_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\cos\left(\theta_{flywheel}(t)\right) = 0. \quad (198)$$

Assuming

$$\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \neq 0, \quad (199)$$

the exploratory equation of motion is written as

$$\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} + \frac{g}{r_{flywheel}}\cos\left(\theta_{flywheel}(t)\right) = 0. \quad (200)$$

### 5.12 Characteristic Angular Velocity

The characteristic angular velocity introduced by the exploratory model is

$$\omega_{flywheel} = \sqrt{\frac{g}{r_{flywheel}}}. \tag{201}$$

This relation will be investigated further in subsequent theoretical and numerical analyses.

### 5.13 Gravitational Energy Balance

The purpose of the gravitational model is not to assume the existence of net energy production.

The role of the present formulation is to track the exchange between
— gravitational potential energy ;
— kinetic energy ;
— rotational energy ;
— electrical energy ;
— dissipative losses.

Any apparent energy surplus observed in simulations must be evaluated using the complete system-level energy balance developed in subsequent chapters.

The equations presented in this chapter therefore constitute a preliminary analytical framework for further investigation rather than a demonstration of net energy generation.

## 6 Rotational Dynamics

The rotational dynamics of the proposed gravitational engine are described using classical rigid-body mechanics. This section establishes the relationship between applied torque, angular acceleration, rotational kinetic energy, and angular momentum.

In addition to the standard equations of rotational mechanics, this chapter incorporates the preliminary analytical equations currently used in the research programme. At this stage, these relations are retained in their present form as working hypotheses and exploratory modelling equations. Their validity will be examined through subsequent analytical development, numerical simulation, dimensional analysis, and complete energy-balance verification.

### 6.1 Angular Position and Angular Velocity

The rotational state of a rigid body is described by its angular position

$$\theta(t). \tag{202}$$

The angular velocity is defined as

$$\omega(t) = \frac{d\theta}{dt}. \tag{203}$$

The angular acceleration is therefore

$$\alpha(t) = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}. \tag{204}$$

These quantities provide the fundamental kinematic variables used to describe the rotating components of the machine.

## 6.2 Fundamental Principle of Rotating Dynamics

The tangential component of the resultant force is written as

$$F_t = ma_t. \tag{205}$$

The moment of force with respect to the origin is written as

$$C = F_t r. \tag{206}$$

The tangential acceleration is written as

$$a_t = r\frac{\mathrm{d}\omega}{\mathrm{d}t}. \tag{207}$$

Combining the previous equations gives

$$C = mr^2\frac{\mathrm{d}\omega}{\mathrm{d}t}. \tag{208}$$

These equations provide an initial representation of the rotational response of the system.

## 6.3 Moment of Inertia

For a rigid body rotating about a fixed axis, the moment of inertia is

$$I = \int r^2\, dm, \tag{209}$$

where $r$ is the perpendicular distance between an elementary mass $dm$ and the axis of rotation.

For a system composed of several rotating components, the total moment of inertia can be expressed as

$$I_{\text{tot}} = \sum_i I_i. \tag{210}$$

If individual components rotate at different angular velocities, their kinetic energies must instead be considered separately.

## 6.4 Rotational Equation of Motion

The fundamental equation governing rotational motion is

$$\sum \tau = I\alpha, \tag{211}$$

where $\tau$ represents the applied torque.

For the gravitational engine model, the net torque can be decomposed into the contributions of gravitational, motor, generator, frictional, and other mechanical torques :

$$\tau_{\text{net}} = \tau_{\text{grav}} + \tau_{\text{motor}} - \tau_{\text{generator}} - \tau_{\text{friction}} + \tau_{\text{other}}. \tag{212}$$

Consequently,

$$I_{\text{tot}}\alpha = \tau_{\text{net}}. \tag{213}$$

This equation provides the basic rotational dynamic equation used by the subsequent machine model.

### 6.5 Gravitational Torque

When a mass $m$ is subjected to gravity, the gravitational force is

$$F_g = mg, \tag{214}$$

where $g$ denotes the local gravitational acceleration.

If the position of the mass relative to the rotational axis is described by a vector $\mathbf{r}$, the gravitational torque is

$$\boldsymbol{\tau}_{\text{grav}} = \mathbf{r} \times m\mathbf{g}. \tag{215}$$

For planar motion, the corresponding scalar torque may be written as

$$\tau_{\text{grav}} = mgr\sin(\theta), \tag{216}$$

with the sign determined by the chosen angular convention.

The gravitational torque is therefore dependent on both the geometry of the mechanism and its instantaneous angular position.

### 6.6 Simple Pendulum Rotational Energy

The gravitational potential energy is written as

$$E_p = mgr\left(1 - \cos(\theta)\right). \tag{217}$$

The kinetic energy is written as

$$E_c = \frac{1}{2}mV^2. \tag{218}$$

Using

$$V = \omega r, \tag{219}$$

one obtains

$$E_c = \frac{1}{2}m\left(\omega r\right)^2. \tag{220}$$

The total mechanical energy is written as

$$E_m = E_c + E_p. \tag{221}$$

Therefore,

$$E_m = \frac{1}{2}m\left(\omega r\right)^2 + mgr\left(1 - \cos(\theta)\right). \tag{222}$$

### 6.7 Rotational Kinetic Energy

The kinetic energy associated with a rotating rigid body is

$$E_{\text{rot}} = \frac{1}{2}I\omega^2. \tag{223}$$

For multiple rotating bodies,

$$E_{\text{rot,tot}} = \sum_i \frac{1}{2}I_i\omega_i^2. \tag{224}$$

This quantity is important when analysing energy storage in rotating components such as flywheels.

The time derivative of rotational kinetic energy is related to mechanical power :

$$P_{\mathrm{rot}} = \frac{dE_{\mathrm{rot}}}{dt}. \tag{225}$$

Using the rotational equation of motion,

$$P_{\mathrm{rot}} = \tau\omega. \tag{226}$$

This relationship provides a direct connection between torque, angular velocity, and mechanical power.

## 6.8 Preliminary Flywheel Model

The flywheel is represented by

$$M_{flywheel}, \tag{227}$$

$$r_{flywheel}, \tag{228}$$

and

$$\theta_{flywheel}(t). \tag{229}$$

The angular velocity of the flywheel is

$$\omega_{flywheel} = \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}. \tag{230}$$

The kinetic energy of the flywheel is written as

$$E_{c,flywheel} = M_{flywheel} \left(\omega_{flywheel} r_{flywheel}\right)^2. \tag{231}$$

The gravitational potential energy is written as

$$E_{p,flywheel} = 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{232}$$

The mechanical energy of the flywheel is written as

$$E_{m,flywheel} = M_{flywheel} \left(\omega_{flywheel} r_{flywheel}\right)^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{233}$$

Using

$$\omega_{flywheel} = \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}, \tag{234}$$

the mechanical energy becomes

$$E_{m,flywheel} = M_{flywheel} \left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\right)^2 r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{235}$$

The exploratory model assumes

$$E_{m,flywheel} = constant. \tag{236}$$

Consequently,

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 0. \tag{237}$$

Differentiating the previous expression gives

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 2M_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2}r_{flywheel}^2 + 2M_{flywheel}g r_{flywheel}\cos\left(\theta_{flywheel}(t)\right). \tag{238}$$

The expanded equation is written as

$$2M_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2}r_{flywheel}^2 + 2M_{flywheel}g r_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\cos\left(\theta_{flywheel}(t)\right) = 0. \tag{239}$$

Assuming

$$\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \neq 0, \tag{240}$$

the exploratory equation of motion is written as

$$\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} + \frac{g}{r_{flywheel}}\cos\left(\theta_{flywheel}(t)\right) = 0. \tag{241}$$

The characteristic angular velocity is introduced as

$$\omega_{flywheel} = \sqrt{\frac{g}{r_{flywheel}}}. \tag{242}$$

## 6.9 Angular Momentum

The angular momentum of a rigid body rotating about a fixed axis is

$$L = I\omega. \tag{243}$$

The corresponding dynamic equation is

$$\tau_{\mathrm{net}} = \frac{dL}{dt}. \tag{244}$$

For constant moment of inertia,

$$\tau_{\mathrm{net}} = I\frac{d\omega}{dt}. \tag{245}$$

If the moment of inertia varies with time, the more general expression is

$$\tau_{\mathrm{net}} = I\frac{d\omega}{dt} + \omega\frac{dI}{dt}. \tag{246}$$

The latter formulation may become relevant if the proposed mechanism involves changes in mass distribution during operation.

## 6.10 Mechanical Losses

A realistic model must account for mechanical losses. A first-order representation of viscous rotational losses is

$$\tau_{\text{friction}} = c\omega, \tag{247}$$

where $c$ is an effective viscous damping coefficient.

A more general model may include Coulomb friction :

$$\tau_{\text{friction}} = c\omega + \tau_c \operatorname{sgn}(\omega), \tag{248}$$

where $\tau_c$ represents the Coulomb friction torque.

The inclusion of these terms is necessary when comparing an ideal theoretical model with a physically realizable machine.

## 6.11 Motor and Generator Rotational Quantities

The motor power is written as

$$P_{motor} = C_{motor}\omega_{motor}. \tag{249}$$

The generator rotational speed is introduced as

$$\omega_{generator} = 60 f_{generator}. \tag{250}$$

For

$$f_{generator} = 50, \tag{251}$$

one obtains

$$\omega_{generator} = 3000. \tag{252}$$

The pulley-belt transmission ratio is written as

$$\frac{d_{out}}{d_{in}} = \frac{\omega_{in}}{\omega_{out}} = \frac{C_{out}}{C_{in}}. \tag{253}$$

## 6.12 Rotational Power Balance

The instantaneous mechanical power associated with a torque is

$$P = \tau\omega. \tag{254}$$

Accordingly, the rotational power balance can be expressed in the form

$$P_{\text{net}} = P_{\text{grav}} + P_{\text{motor}} - P_{\text{generator}} - P_{\text{loss}}. \tag{255}$$

where each term represents the instantaneous mechanical power associated with the corresponding torque.

The change in stored rotational energy is then

$$\frac{dE_{\text{rot}}}{dt} = P_{\text{net}}. \tag{256}$$

This equation is particularly important for evaluating whether an apparent increase in mechanical output corresponds to a genuine external energy input or merely to a temporary release of previously stored energy.

### 6.13 Periodic Operation

If the gravitational engine operates periodically, the rotational state must be examined over a complete cycle rather than at a single instant.

For a periodic steady state with period $T$, the angular velocity must satisfy

$$\omega(t+T) = \omega(t). \tag{257}$$

Likewise, the mechanical state of the complete system must return to its initial state after one complete cycle, subject to any explicitly defined external energy or mass exchanges.

The net mechanical work performed over one cycle is

$$W_{\text{cycle}} = \int_0^T \tau_{\text{net}}(t)\omega(t)\,dt. \tag{258}$$

Since

$$P(t) = \tau(t)\omega(t), \tag{259}$$

the cycle work may equivalently be written as

$$W_{\text{cycle}} = \int_0^T P(t)\,dt. \tag{260}$$

This cycle-integrated formulation will be used later to distinguish transient energy storage from sustained mechanical energy production.

### 6.14 Role in the Overall Model

The rotational dynamics developed in this section form the mechanical core of the proposed system model. They provide the equations required to connect gravitational forces, rotating masses, flywheel behaviour, motor-generator interaction, and mechanical losses.

The model does not assume that gravitational torque constitutes a continuous source of net energy. Instead, the gravitational interaction is treated as a conservative mechanical interaction unless an explicitly defined external energy or mass input is introduced.

Consequently, the complete system must satisfy an energy balance over each operating cycle. This condition will be used in the subsequent sections to evaluate the proposed machine quantitatively and to identify whether any predicted mechanical output can be attributed to an external energy source, stored energy, or modelling assumptions.

## 7 Flywheel Model

The flywheel is modelled as a rotating energy-storage element within the proposed gravitational engine. Its primary function in the theoretical model is to store and release rotational kinetic energy while maintaining a defined angular state.

In addition to the standard rigid-body flywheel representation, this chapter incorporates the preliminary analytical equations currently used within the research project. These equations are retained in their present form as working hypotheses and exploratory model relationships. Their physical validity and applicability will be examined in subsequent chapters through analytical derivations, numerical simulations, and complete energy-balance verification.

## 7.1 Flywheel Geometry

For a rigid flywheel rotating about a fixed axis, the moment of inertia is

$$I_f = \int r^2 \, dm. \tag{261}$$

For an ideal thin ring of mass $M_f$ and radius $R_f$,

$$I_f = M_f R_f^2. \tag{262}$$

For a uniform solid disk,

$$I_f = \frac{1}{2} M_f R_f^2. \tag{263}$$

The actual value of $I_f$ must be determined from the geometry and mass distribution of the physical flywheel.

## 7.2 Angular State

The angular position of the flywheel is denoted by

$$\theta_f(t). \tag{264}$$

Its angular velocity is

$$\omega_f(t) = \frac{d\theta_f}{dt}, \tag{265}$$

and its angular acceleration is

$$\alpha_f(t) = \frac{d\omega_f}{dt}. \tag{266}$$

For a flywheel with constant moment of inertia, the rotational equation of motion is

$$I_f \frac{d\omega_f}{dt} = \tau_f, \tag{267}$$

where $\tau_f$ represents the net torque applied to the flywheel.

## 7.3 Stored Rotational Energy

The rotational kinetic energy stored in the flywheel is

$$E_f = \frac{1}{2} I_f \omega_f^2. \tag{268}$$

This equation is fundamental to the flywheel model because it establishes the maximum amount of mechanical energy that can be temporarily stored for a given moment of inertia and angular velocity.

The change in stored energy between two angular velocities is

$$\Delta E_f = \frac{1}{2} I_f \left( \omega_{f,2}^2 - \omega_{f,1}^2 \right). \tag{269}$$

An increase in flywheel speed therefore corresponds to an increase in stored mechanical energy, while a decrease in speed releases previously stored energy.

### 7.4 Preliminary Flywheel Parameters

The exploratory flywheel model introduces the following parameters.
The flywheel mass is

$$M_{flywheel}. \tag{270}$$

The flywheel radius is

$$r_{flywheel}. \tag{271}$$

The angular position is

$$\theta_{flywheel}(t). \tag{272}$$

The angular velocity is

$$\omega_{flywheel} = \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}. \tag{273}$$

### 7.5 Preliminary Flywheel Kinetic Energy

The kinetic energy of the flywheel is written as

$$E_{c,flywheel} = M_{flywheel} \left(\omega_{flywheel} r_{flywheel}\right)^2. \tag{274}$$

This expression is retained in its current form as part of the preliminary analytical model.

### 7.6 Preliminary Flywheel Potential Energy

The gravitational potential energy is written as

$$E_{p,flywheel} = 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{275}$$

This relation is currently used as the preliminary gravitational-energy representation associated with the flywheel.

### 7.7 Preliminary Flywheel Mechanical Energy

The total mechanical energy is written as

$$E_{m,flywheel} = M_{flywheel} \left(\omega_{flywheel} r_{flywheel}\right)^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{276}$$

Using

$$\omega_{flywheel} = \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}, \tag{277}$$

the mechanical energy can be written as

$$E_{m,flywheel} = M_{flywheel} \left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} r_{flywheel}\right)^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{278}$$

or

$$E_{m,flywheel} = M_{flywheel} \left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\right)^2 r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{279}$$

### 7.8 Preliminary Conservation Assumption

The exploratory model assumes

$$E_{m,flywheel} = constant. \tag{280}$$

Consequently,

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 0. \tag{281}$$

Differentiating the previous expression gives

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 2M_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2}r_{flywheel}^2 + 2M_{flywheel}g r_{flywheel}\cos\left(\theta_{flywheel}(t)\right). \tag{282}$$

The expanded expression is written as

$$2M_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2}r_{flywheel}^2 + 2M_{flywheel}g r_{flywheel}\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\cos\left(\theta_{flywheel}(t)\right) = 0. \tag{283}$$

Assuming

$$\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \neq 0, \tag{284}$$

the exploratory equation of motion is written as

$$\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} + \frac{g}{r_{flywheel}}\cos\left(\theta_{flywheel}(t)\right) = 0. \tag{285}$$

The characteristic angular velocity is introduced as

$$\omega_{flywheel} = \sqrt{\frac{g}{r_{flywheel}}}. \tag{286}$$

### 7.9 Flywheel Torque

The torque required to accelerate the flywheel is

$$\tau_f = I_f \alpha_f. \tag{287}$$

Using

$$\alpha_f = \frac{d\omega_f}{dt}, \tag{288}$$

the equation becomes

$$\tau_f = I_f \frac{d\omega_f}{dt}. \tag{289}$$

The mechanical power transferred to or from the flywheel is

$$P_f = \tau_f \omega_f. \tag{290}$$

Consequently,

$$P_f = I_f \omega_f \frac{d\omega_f}{dt}. \tag{291}$$

This is equivalent to the time derivative of the rotational kinetic energy :

$$P_f = \frac{dE_f}{dt}. \tag{292}$$

## 7.10 Flywheel Acceleration and Deceleration

During an acceleration phase, the flywheel receives positive mechanical power and its stored kinetic energy increases :

$$\frac{dE_f}{dt} > 0. \tag{293}$$

During a deceleration phase,

$$\frac{dE_f}{dt} < 0, \tag{294}$$

and stored rotational energy is transferred to another component of the system.
The energy released during deceleration from $\omega_{\text{high}}$ to $\omega_{\text{low}}$ is

$$E_{\text{released}} = \frac{1}{2} I_f \left(\omega_{\text{high}}^2 - \omega_{\text{low}}^2\right). \tag{295}$$

This quantity represents an energy transfer from the flywheel rather than a new source of energy.

## 7.11 Mechanical Losses

A realistic flywheel model must include mechanical losses. A simple viscous-loss model is

$$\tau_{\text{loss}} = c_f \omega_f, \tag{296}$$

where $c_f$ is an effective rotational damping coefficient.
The corresponding power loss is

$$P_{\text{loss}} = \tau_{\text{loss}} \omega_f = c_f \omega_f^2. \tag{297}$$

A model including Coulomb friction may additionally be written as

$$\tau_{\text{loss}} = c_f \omega_f + \tau_c \operatorname{sgn}(\omega_f), \tag{298}$$

where $\tau_c$ is the Coulomb friction torque.
The choice of loss model can be refined when experimental data become available.

## 7.12 Motor and Generator Relationships

The motor power is written as

$$P_{motor} = C_{motor} \omega_{motor}. \tag{299}$$

The motor angular speed is introduced as

$$\omega_{motor} = 1500. \tag{300}$$

The motor torque is written as

$$C_{motor} = 2 M_{flywheel} g r_{flywheel}. \tag{301}$$

The mechanical power is written as

$$P_{m,motor} = C_{motor}\omega_{motor}. \tag{302}$$

Consequently,

$$P_{m,motor} = 2M_{flywheel}gr_{flywheel}\omega_{motor}. \tag{303}$$

If there is no resistance,

$$P_{e,motor} = P_{m,motor}. \tag{304}$$

Therefore,

$$P_{e,motor} = 2M_{flywheel}gr_{flywheel}\omega_{motor}. \tag{305}$$

The generator speed is written as

$$\omega_{generator} = 60f_{generator}. \tag{306}$$

For

$$f_{generator} = 50, \tag{307}$$

one obtains

$$\omega_{generator} = 3000. \tag{308}$$

The electrical power is written as

$$P_{e,generator} = E_{e,generator}\omega_{generator}. \tag{309}$$

Equivalently,

$$E_{e,generator} = \frac{P_{e,generator}}{\omega_{generator}}. \tag{310}$$

The preliminary model assumes

$$E_{e,generator} = E_{m,flywheel}. \tag{311}$$

Therefore,

$$M_{flywheel}\left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\right)^2 r_{flywheel}^2 + 2M_{flywheel}gr_{flywheel}\sin\left(\theta_{flywheel}(t)\right) = \frac{P_{e,generator}}{\omega_{generator}}. \tag{312}$$

### 7.13 Pulley Relationships

Let

$$d_{pulley,motor} \tag{313}$$

be the motor pulley diameter.
Let

$$d_{pulley,generator} \tag{314}$$

be the generator pulley diameter.

The flywheel angular speed is written as

$$\omega_{flywheel} = \frac{d_{pulley,motor}\omega_{motor}}{d_{pulley,flywheel}}. \tag{315}$$

The flywheel pulley diameter is written as

$$d_{pulley,flywheel} = \frac{\omega_{generator}d_{pulley,generator}}{\omega_{flywheel}}. \tag{316}$$

The generator pulley diameter is written as

$$d_{pulley,generator} = \frac{\omega_{flywheel}d_{pulley,flywheel}}{\omega_{generator}}. \tag{317}$$

The motor pulley diameter is written as

$$d_{pulley,motor} = \frac{\omega_{flywheel}d_{pulley,flywheel}}{\omega_{motor}}. \tag{318}$$

With

$$\theta_{flywheel}(t) = \frac{\pi}{2}, \tag{319}$$

the flywheel mass is written as

$$M_{flywheel} = \frac{P_{e,generator}}{2\omega_{generator}g r_{flywheel}}. \tag{320}$$

### 7.14 Complete Flywheel Dynamic Equation

When the flywheel is coupled to several mechanical elements, its net torque can be expressed as

$$\tau_{\text{net},f} = \tau_{\text{input},f} - \tau_{\text{output},f} - \tau_{\text{loss},f}. \tag{321}$$

The corresponding equation of motion is

$$I_f \frac{d\omega_f}{dt} = \tau_{\text{input},f} - \tau_{\text{output},f} - \tau_{\text{loss},f}. \tag{322}$$

This equation provides the principal state equation for the flywheel in the numerical model.

### 7.15 Energy Balance

The flywheel does not create mechanical energy. It acts as an energy storage component.

For an interval $[t_1, t_2]$, the energy balance can be expressed as

$$E_f(t_2) - E_f(t_1) = \int_{t_1}^{t_2} P_{\text{input},f}\, dt - \int_{t_1}^{t_2} P_{\text{output},f}\, dt - \int_{t_1}^{t_2} P_{\text{loss},f}\, dt. \tag{323}$$

This relationship is an important validation condition for the simulation model.

In particular, if the flywheel returns to its initial angular velocity after a complete operating cycle,

$$E_f(T) = E_f(0), \tag{324}$$

then the net change in stored flywheel energy over that cycle is zero :

$$\Delta E_{f,\text{cycle}} = 0. \tag{325}$$

The flywheel can therefore redistribute energy within a cycle, but it cannot provide a persistent net energy surplus without an external energy input.

## 7.16 Periodic Operation

For a periodic operating regime with period $T$, the flywheel state is expected to satisfy

$$\omega_f(t+T) = \omega_f(t) \tag{326}$$

when the system has reached periodic steady state.

The corresponding cycle-integrated energy balance is

$$\int_0^T P_{\text{input},f}\, dt = \int_0^T P_{\text{output},f}\, dt + \int_0^T P_{\text{loss},f}\, dt. \tag{327}$$

This condition provides a useful diagnostic for the numerical simulations developed later in the research.

## 7.17 Flywheel as an Energy Buffer

The flywheel can nevertheless play an important engineering role in the proposed system.

Because the instantaneous mechanical power associated with the gravitational and electromechanical subsystems may vary with time, the flywheel can act as a buffer between periods of energy input and periods of energy extraction.

The flywheel can therefore :

— absorb temporary mechanical power ;
— release previously stored mechanical energy ;
— reduce short-term angular velocity variations ;
— smooth transient torque variations ;
— provide a well-defined rotational state for the motor-generator subsystem.

These functions are compatible with conventional mechanical energy storage and do not imply net energy generation.

## 7.18 Model Parameters

The numerical implementation of the flywheel model requires at least the following parameters :

$$\mathcal{P}_f = \{M_f,\, R_f,\, I_f,\, c_f,\, \tau_c,\, \omega_{f,0}\}\,. \tag{328}$$

Here $M_f$ is the flywheel mass, $R_f$ is a characteristic radius, $I_f$ is the moment of inertia, $c_f$ is the viscous damping coefficient, $\tau_c$ is the Coulomb friction torque, and $\omega_{f,0}$ is the initial angular velocity.

When sufficient geometric information is available, $I_f$ should be calculated directly from the physical mass distribution rather than treated as an independent fitted parameter.

## 7.19 Role in the Overall Machine Model

The flywheel model provides the mechanical energy-storage component required for coupling the rotational dynamics to the motor-generator and gravitational subsystems.

In the complete machine model, the flywheel state variables will be combined with the angular positions and velocities of the other rotating components.

The resulting coupled system will allow the research to evaluate transient behaviour, energy transfer, mechanical losses, and periodic steady-state operation while maintaining an explicit distinction between stored energy and externally supplied energy.

# 8 Motor and Generator Model

The motor-generator subsystem provides the electromechanical interface between the rotating mechanical system and the electrical domain. In the present theoretical model, the motor and generator are treated as reversible electromechanical devices whose operating mode depends on the direction of energy transfer.

The purpose of this section is to establish a mathematical representation suitable for coupling the electrical and mechanical models developed in the subsequent analysis.

In addition to the standard electromechanical equations used in rotating electrical machines, this chapter incorporates the preliminary analytical equations currently used in the research programme. At this stage, these equations are retained in their present form as exploratory modelling relationships and working hypotheses. Their validity will be examined through analytical derivations, numerical simulations, dimensional analysis, and complete energy-balance verification.

## 8.1 Electromechanical Conversion

An ideal rotating electrical machine converts mechanical power into electrical power when operating as a generator, and electrical power into mechanical power when operating as a motor.

The instantaneous mechanical power is

$$P_{\text{mech}} = \tau_e \omega, \tag{329}$$

where $\tau_e$ is the electromagnetic torque and $\omega$ is the angular velocity of the rotor.

For an ideal lossless conversion,

$$P_{\text{elec}} = P_{\text{mech}}. \tag{330}$$

A realistic machine includes conversion losses. Its efficiency can therefore be represented by

$$\eta_{\text{em}} = \frac{P_{\text{out}}}{P_{\text{in}}}, \tag{331}$$

with the appropriate definition of input and output power depending on whether the machine is operating as a motor or as a generator.

## 8.2 Motor Model

In motor operation, electrical power is supplied to the machine and converted into mechanical power.

The mechanical equation of motion can be written as

$$I_m \frac{d\omega}{dt} = \tau_m - \tau_{\text{load}} - \tau_{\text{loss}}, \tag{332}$$

where $I_m$ is the equivalent rotor inertia, $\tau_m$ is the electromagnetic motor torque, $\tau_{\text{load}}$ is the external mechanical load torque, and $\tau_{\text{loss}}$ represents mechanical losses.

A simplified torque relationship may be expressed as

$$\tau_m = K_t i, \tag{333}$$

where $K_t$ is the torque constant and $i$ is the relevant motor current.

The corresponding mechanical power is

$$P_m = \tau_m \omega. \tag{334}$$

## 8.3 Preliminary Motor Parameters

The angular speed of the motor is introduced as

$$\omega_{motor} = 1500. \tag{335}$$

The preliminary motor torque is written as

$$C_{motor} = 2M_{flywheel} g r_{flywheel}. \tag{336}$$

The mechanical power is written as

$$P_{m,motor} = C_{motor}\omega_{motor}. \tag{337}$$

Consequently,

$$P_{m,motor} = 2M_{flywheel} g r_{flywheel}\omega_{motor}. \tag{338}$$

If there is no resistance, the electrical power is written as

$$P_{e,motor} = P_{m,motor}. \tag{339}$$

Thus,

$$P_{e,motor} = 2M_{flywheel} g r_{flywheel}\omega_{motor}. \tag{340}$$

Let

$$d_{pulley,motor} \tag{341}$$

denote the diameter of the motor pulley in metres.

## 8.4 Generator Model

In generator operation, mechanical energy is supplied to the rotating machine and converted into electrical energy.

The electromagnetic torque produced by the generator opposes the mechanical rotation under normal power-generation operation. The mechanical equation can therefore be written as

$$I_g \frac{d\omega}{dt} = \tau_{\text{drive}} - \tau_g - \tau_{\text{loss}}, \tag{342}$$

where $I_g$ is the generator inertia, $\tau_{\text{drive}}$ is the driving mechanical torque, and $\tau_g$ is the electromagnetic generator torque.

The electrical output power is related to the generator torque by

$$P_g = \tau_g\omega. \tag{343}$$

In the ideal case,

$$P_{\text{electrical,out}} = P_g. \tag{344}$$

For a real machine,

$$P_{\text{electrical,out}} = \eta_g \tau_g \omega, \tag{345}$$

where $\eta_g$ is the generator conversion efficiency.

## 8.5 Preliminary Generator Parameters

The generator speed is written as

$$\omega_{generator} = 60 f_{generator}. \tag{346}$$

For

$$f_{generator} = 50, \tag{347}$$

the preliminary model gives

$$\omega_{generator} = 60 \times 50. \tag{348}$$

Therefore,

$$\omega_{generator} = 3000. \tag{349}$$

The electrical power is written as

$$P_{e,generator} = E_{e,generator} \omega_{generator}. \tag{350}$$

Equivalently,

$$E_{e,generator} = \frac{P_{e,generator}}{\omega_{generator}}. \tag{351}$$

The exploratory model further assumes

$$E_{e,generator} = E_{m,flywheel}. \tag{352}$$

Using the current flywheel model,

$$M_{flywheel} \left( \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \right)^2 r_{flywheel}^2 + 2M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right) = \frac{P_{e,generator}}{\omega_{generator}}. \tag{353}$$

Let

$$d_{pulley,generator} \tag{354}$$

denote the diameter of the generator pulley in metres.

## 8.6 Back Electromotive Force

For a simplified rotating electrical machine, the generated electromotive force can be represented as

$$E = K_e \omega, \tag{355}$$

where $K_e$ is the back-electromotive-force constant.

A simple electrical circuit model is

$$V = Ri + L\frac{di}{dt} + K_e \omega, \tag{356}$$

where $V$ is the applied electrical voltage, $R$ is the winding resistance, and $L$ is the effective winding inductance.

The exact form of the electrical equation depends on the machine topology. The above equation is used only as a first-order lumped representation.

## 8.7 Torque and Current

For a simplified linear electromechanical model, electromagnetic torque is proportional to current :

$$\tau_e = K_t i. \tag{357}$$

The electromagnetic power is therefore

$$P_e = \tau_e \omega = K_t i \omega. \tag{358}$$

Under a consistent choice of SI units and machine constants, the idealized reciprocal relationship between $K_t$ and $K_e$ provides the electromechanical coupling between the electrical and mechanical domains.

## 8.8 Electrical Losses

The primary electrical loss considered in the initial model is resistive loss :

$$P_R = Ri^2. \tag{359}$$

Additional losses may include magnetic losses, switching losses, mechanical bearing losses, windage, and other parasitic effects.

For the initial theoretical model, these additional losses may be represented by an aggregate term

$$P_{\text{loss,elec}} = P_R + P_{\text{mag}} + P_{\text{other}}. \tag{360}$$

The parameters associated with these losses should be determined from machine specifications or experimental measurements whenever possible.

## 8.9 Mechanical Losses

The mechanical losses of the rotating electrical machine may be represented using a viscous damping model :

$$\tau_{\text{visc}} = c_m \omega. \tag{361}$$

The associated power loss is

$$P_{\text{visc}} = c_m \omega^2. \tag{362}$$

A Coulomb friction contribution can additionally be included :

$$\tau_{\text{friction}} = c_m \omega + \tau_c \operatorname{sgn}(\omega). \tag{363}$$

The complete loss model can be refined as experimental data become available.

## 8.10 Pulley and Transmission Relationships

The pulley-belt transmission ratio is written as

$$\frac{d_{out}}{d_{in}} = \frac{\omega_{in}}{\omega_{out}} = \frac{C_{out}}{C_{in}}. \tag{364}$$

The flywheel angular speed is written as

$$\omega_{flywheel} = \frac{d_{pulley,motor}\omega_{motor}}{d_{pulley,flywheel}}. \tag{365}$$

The flywheel pulley diameter is written as

$$d_{pulley,flywheel} = \frac{\omega_{generator} d_{pulley,generator}}{\omega_{flywheel}}. \tag{366}$$

The generator pulley diameter is written as

$$d_{pulley,generator} = \frac{\omega_{flywheel} d_{pulley,flywheel}}{\omega_{generator}}. \tag{367}$$

The motor pulley diameter is written as

$$d_{pulley,motor} = \frac{\omega_{flywheel} d_{pulley,flywheel}}{\omega_{motor}}. \tag{368}$$

## 8.11 Motor-Generator Energy Balance

For the electromechanical subsystem, conservation of energy requires

$$P_{\text{electrical,in}} + P_{\text{mechanical,in}} = P_{\text{electrical,out}} + P_{\text{mechanical,out}} + P_{\text{loss}}. \tag{369}$$

For motor operation, the principal energy flow is

$$P_{\text{electrical,in}} \to P_{\text{mechanical,out}}. \tag{370}$$

For generator operation, the direction is reversed :

$$P_{\text{mechanical,in}} \to P_{\text{electrical,out}}. \tag{371}$$

The machine therefore transfers energy between domains but does not constitute an independent source of energy.

## 8.12 Motor-Generator Coupling

When the motor-generator is mechanically coupled to the flywheel, the combined rotational equation may be written as

$$I_{\text{eq}} \frac{d\omega}{dt} = \tau_{\text{grav}} + \tau_{\text{motor}} - \tau_{\text{generator}} - \tau_{\text{loss}}, \tag{372}$$

where $I_{\text{eq}}$ is the equivalent rotational inertia referred to the common shaft.

The corresponding mechanical power balance is

$$\frac{dE_{\text{rot}}}{dt} = P_{\text{grav}} + P_{\text{motor}} - P_{\text{generator}} - P_{\text{loss}}. \tag{373}$$

This equation provides the principal coupling between the rotational dynamics and the electrical subsystem.

## 8.13 Operating Modes

The machine model distinguishes between several possible operating conditions.

- **Motor mode :** electrical energy is supplied to produce mechanical torque.
- **Generator mode :** mechanical energy is converted into electrical energy.
- **Regenerative mode :** mechanical energy stored in the rotating system is returned to the electrical subsystem.
- **Idle mode :** the machine exchanges negligible useful power with the external electrical system.

The transition between these modes must be explicitly defined in a numerical simulation to avoid introducing artificial energy into the system.

## 8.14 Control Variables

The principal electrical control variables may include voltage, current, electromagnetic torque, and generator load.

A generic control input can be represented by

$$u(t) = [V(t),\, i(t),\, \tau_e(t)]\,. \tag{374}$$

The control strategy must be defined independently from the physical energy balance. In particular, a controller can redistribute energy within the system but cannot create net energy.

## 8.15 Efficiency

For motor operation, the mechanical output efficiency may be written as

$$\eta_m = \frac{P_{\text{mechanical,out}}}{P_{\text{electrical,in}}}. \tag{375}$$

For generator operation,

$$\eta_g = \frac{P_{\text{electrical,out}}}{P_{\text{mechanical,in}}}. \tag{376}$$

For a motor-generator cycle involving both conversion directions, the combined efficiency is approximately

$$\eta_{\text{cycle}} = \eta_m \eta_g, \tag{377}$$

provided that the two conversion efficiencies are defined consistently and other losses are treated separately.

Since realistic efficiencies satisfy

$$0 < \eta_m < 1, \qquad 0 < \eta_g < 1, \tag{378}$$

a complete electrical-mechanical-electrical conversion cycle necessarily involves energy losses.

## 8.16 Cycle Energy Balance

For a complete operating cycle of duration $T$, the net electrical energy delivered externally is

$$E_{\text{out}} = \int_0^T P_{\text{electrical,out}}(t)\, dt. \tag{379}$$

The corresponding electrical energy supplied to the machine is

$$E_{\text{in}} = \int_0^T P_{\text{electrical,in}}(t)\, dt. \tag{380}$$

The complete system must also account for changes in stored mechanical energy :

$$\Delta E_{\text{stored}} = E_{\text{in}} - E_{\text{out}} - E_{\text{loss}}, \tag{381}$$

where $E_{\text{loss}}$ represents the total dissipated energy over the cycle.

For a periodic steady state in which all internal state variables return to their initial values,

$$\Delta E_{\text{stored}} = 0. \tag{382}$$

The resulting balance becomes

$$E_{\text{in}} = E_{\text{out}} + E_{\text{loss}}. \tag{383}$$

This condition is an essential validation criterion for the numerical model.

### 8.17 Exploratory Flywheel-Generator Relationship

With

$$\theta_{flywheel}(t) = \frac{\pi}{2}, \tag{384}$$

the flywheel mass is written as

$$M_{flywheel} = \frac{P_{e,generator}}{2\omega_{generator} g r_{flywheel}}. \tag{385}$$

This relationship is retained in its current form as part of the preliminary analytical model and will be evaluated in later stages of the research.

### 8.18 Role in the Complete Machine Model

The motor-generator model provides the electromechanical coupling required to connect the rotating gravitational and flywheel subsystems to an electrical load.

The subsequent machine model will combine the equations established in the previous sections into a unified dynamical system. This will allow the simulation to track angular position, angular velocity, torque, electrical current, electrical power, stored energy, and dissipative losses simultaneously.

The central modelling principle is that the motor-generator subsystem can transfer and convert energy, but cannot provide a net energy source without an explicitly defined external input.

## 9 Machine Model

This chapter defines the global mathematical model of the Chas Campbell Gravitational Engine as an interconnected electromechanical system. The objective is to establish a coherent system-level representation that can subsequently be implemented numerically and evaluated through simulation.

The standard system equations and the author's current exploratory equations are retained together. The latter are identified as preliminary working relationships whose dimensional consistency, physical validity, and applicability remain to be evaluated without presupposing net energy production.

### 9.1 System Architecture

The machine is considered as a coupled system composed of the following principal subsystems :

— a gravitational subsystem ;
— a rotating mechanical assembly ;
— one or more flywheels ;
— an electric motor ;
— an electric generator ;
— mechanical transmission elements ;
— structural constraints and bearings ;
— control and measurement elements.

The complete system is represented schematically as an energy and momentum transfer chain :

$$\text{Electrical Input} \rightarrow \text{Motor} \rightarrow \text{Mechanical Assembly} \rightarrow \text{Generator} \rightarrow \text{Electrical Output}. \quad (386)$$

The gravitational and rotational subsystems are coupled through the geometry and motion of the masses and rotating components.

## 9.2 Generalized Coordinates

The configuration of the machine is described using a set of generalized coordinates

$$\mathbf{q} = \begin{bmatrix} q_1 & q_2 & \dots & q_n \end{bmatrix}^T. \quad (387)$$

The corresponding generalized velocities and accelerations are

$$\dot{\mathbf{q}} = \frac{d\mathbf{q}}{dt}, \qquad \ddot{\mathbf{q}} = \frac{d^2\mathbf{q}}{dt^2}. \quad (388)$$

For a simplified single-axis representation, the principal rotational coordinate may be written as

$$\theta(t), \quad (389)$$

with angular velocity

$$\omega(t) = \dot{\theta}(t) \quad (390)$$

and angular acceleration

$$\alpha(t) = \dot{\omega}(t). \quad (391)$$

## 9.3 Mechanical Dynamics

The rotational dynamics of the complete mechanical assembly are governed by conservation of angular momentum. For a single equivalent rotational degree of freedom,

$$I_{\text{eq}}\dot{\omega} = \tau_{\text{motor}} + \tau_{\text{grav}} - \tau_{\text{gen}} - \tau_{\text{loss}}, \quad (392)$$

where

$$I_{\text{eq}} : \text{equivalent rotational inertia}, \quad (393)$$
$$\tau_{\text{motor}} : \text{motor torque}, \quad (394)$$
$$\tau_{\text{grav}} : \text{gravitational torque}, \quad (395)$$
$$\tau_{\text{gen}} : \text{generator reaction torque}, \quad (396)$$
$$\tau_{\text{loss}} : \text{mechanical loss torque}. \quad (397)$$

The equivalent inertia may include contributions from several rotating components :

$$I_{\text{eq}} = I_{\text{rot}} + I_{\text{fw}} + I_{\text{motor}} + I_{\text{gen}} + I_{\text{trans}}, \quad (398)$$

where the individual terms represent the inertia of the principal rotating assembly, flywheel, motor rotor, generator rotor, and transmission components, respectively. When transmission ratios are present, the inertias must be referred to a common rotational axis before being combined.

### 9.4 Gravitational Contribution

For a point mass $m_i$ located at position $\mathbf{r}_i(\mathbf{q})$, the gravitational force is

$$\mathbf{F}_{g,i} = m_i \mathbf{g}, \tag{399}$$

and the corresponding torque about the reference axis is

$$\boldsymbol{\tau}_{g,i} = \mathbf{r}_i \times \mathbf{F}_{g,i}. \tag{400}$$

The total gravitational torque is therefore

$$\boldsymbol{\tau}_{\text{grav}} = \sum_i \mathbf{r}_i \times m_i \mathbf{g}. \tag{401}$$

For a planar rotational model this expression can be reduced to a scalar torque about the principal axis. The gravitational contribution is a function of the machine configuration :

$$\tau_{\text{grav}} = \tau_{\text{grav}}(\mathbf{q}). \tag{402}$$

### 9.5 Preliminary Mechanical Relations

The tangential component of the resultant forces is written as

$$F_t = m a_t. \tag{403}$$

The moment of force with respect to the origin is written as

$$C = F_t r. \tag{404}$$

The tangential acceleration is written as

$$a_t = r \frac{\mathrm{d}\omega}{\mathrm{d}t}. \tag{405}$$

The gravitational potential energy of the simple-pendulum representation is written as

$$E_p = mgr\left(1 - \cos(\theta)\right). \tag{406}$$

The kinetic energy is written as

$$E_c = \frac{1}{2} m V^2 \tag{407}$$

and

$$E_c = \frac{1}{2} m \left(\omega r\right)^2. \tag{408}$$

The mechanical energy is written as

$$E_m = E_c + E_p \tag{409}$$

and consequently

$$E_m = \frac{1}{2} m \left(\omega r\right)^2 + mgr\left(1 - \cos(\theta)\right). \tag{410}$$

## 9.6 Motor Model

The electric motor provides mechanical torque to the rotating system. In the most general representation,

$$\tau_{\text{motor}} = \tau_{\text{motor}}(u, \omega, \mathbf{x}_m). \tag{411}$$

For a simplified electromechanical model,

$$\tau_{\text{motor}} = K_t i_m. \tag{412}$$

The associated electrical power is

$$P_{\text{motor}} = V_m i_m. \tag{413}$$

The mechanical power delivered by the motor is

$$P_{\text{mech,motor}} = \tau_{\text{motor}} \omega. \tag{414}$$

These quantities are not necessarily identical because electrical, mechanical, and thermal losses may be present.

## 9.7 Preliminary Motor Relations

The angular speed of the motor in revolutions per minute is introduced as

$$\omega_{motor} = 1500. \tag{415}$$

The motor torque is written as

$$C_{motor} = 2M_{flywheel} g r_{flywheel}. \tag{416}$$

The mechanical power is written as

$$P_{m,motor} = C_{motor} \omega_{motor} \tag{417}$$

and

$$P_{m,motor} = 2M_{flywheel} g r_{flywheel} \omega_{motor}. \tag{418}$$

If there is no resistance, the electrical power is written as

$$P_{e,motor} = P_{m,motor} \tag{419}$$

and therefore

$$P_{e,motor} = 2M_{flywheel} g r_{flywheel} \omega_{motor}. \tag{420}$$

Let $d_{pulley,motor}$ denote the diameter of the motor pulley in metres.

## 9.8 Generator Model

The generator extracts mechanical power from the rotating system and converts it into electrical power. The mechanical power extracted is

$$P_{\text{gen,mech}} = \tau_{\text{gen}}\omega. \tag{421}$$

The electrical output power is

$$P_{\text{gen,elec}} = V_g i_g. \tag{422}$$

For an idealized generator,

$$P_{\text{gen,elec}} = P_{\text{gen,mech}}. \tag{423}$$

For a realistic machine,

$$P_{\text{gen,elec}} = \eta_{\text{gen}} P_{\text{gen,mech}}, \tag{424}$$

where

$$0 < \eta_{\text{gen}} \leq 1 \tag{425}$$

is the generator efficiency. The generator reaction torque consequently represents a load on the mechanical system.

## 9.9 Preliminary Generator Relations

The generator angular speed in revolutions per minute is written as

$$\omega_{generator} = 60 f_{generator}. \tag{426}$$

For $f_{generator} = 50$,

$$\omega_{generator} = 60 \times 50 \tag{427}$$

and

$$\omega_{generator} = 3000. \tag{428}$$

The electrical power is written as

$$P_{e,generator} = E_{e,generator}\omega_{generator}. \tag{429}$$

Consequently,

$$E_{e,generator} = \frac{P_{e,generator}}{\omega_{generator}}. \tag{430}$$

The preliminary model assumes

$$E_{e,generator} = E_{m,flywheel}. \tag{431}$$

Let $d_{pulley,generator}$ denote the diameter of the generator pulley in metres.

### 9.10 Flywheel Coupling

The conventional rotational kinetic energy of the flywheel is

$$E_{\mathrm{fw}} = \frac{1}{2} I_{\mathrm{fw}} \omega^2. \tag{432}$$

Its rate of change is

$$\frac{dE_{\mathrm{fw}}}{dt} = I_{\mathrm{fw}} \omega \dot{\omega}. \tag{433}$$

Equivalently,

$$P_{\mathrm{fw}} = \tau_{\mathrm{fw}} \omega. \tag{434}$$

The flywheel provides temporary storage and release of rotational kinetic energy. It does not, by itself, constitute a source of net energy.

### 9.11 Preliminary Flywheel Relations

The kinetic energy of the flywheel is written as

$$E_{c,flywheel} = M_{flywheel} \left(\omega_{flywheel} r_{flywheel}\right)^2 . \tag{435}$$

The potential energy of the flywheel is written as

$$E_{p,flywheel} = 2 M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{436}$$

The mechanical energy is written as

$$E_{m,flywheel} = M_{flywheel} \left(\omega_{flywheel} r_{flywheel}\right)^2 + 2 M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{437}$$

Using the time derivative of the angular position,

$$E_{m,flywheel} = M_{flywheel} \left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\right)^2 r_{flywheel}^2 + 2 M_{flywheel} g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right). \tag{438}$$

The current exploratory model assumes

$$E_{m,flywheel} = constant \tag{439}$$

and

$$\frac{\mathrm{d}E_{m,flywheel}}{\mathrm{d}t} = 0. \tag{440}$$

The expanded preliminary relation is

$$2 M_{flywheel} \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} r_{flywheel}^2 + 2 M_{flywheel} g r_{flywheel} \frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \cos\left(\theta_{flywheel}(t)\right) = 0. \tag{441}$$

With

$$\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t} \neq 0, \tag{442}$$

the exploratory equation of motion is written as

$$\frac{\mathrm{d}^2\theta_{flywheel}(t)}{\mathrm{d}t^2} + \frac{g}{r_{flywheel}} \cos\left(\theta_{flywheel}(t)\right) = 0. \tag{443}$$

The characteristic angular speed is written as

$$\omega_{flywheel} = \sqrt{\frac{g}{r_{flywheel}}}. \tag{444}$$

The preliminary flywheel-generator relationship is

$$M_{flywheel}\left(\frac{\mathrm{d}\theta_{flywheel}(t)}{\mathrm{d}t}\right)^2 r_{flywheel}^2 + 2M_{flywheel}g r_{flywheel} \sin\left(\theta_{flywheel}(t)\right) = \frac{P_{e,generator}}{\omega_{generator}}. \tag{445}$$

## 9.12 Transmission Model

The pulley-belt ratio is written as

$$\frac{d_{out}}{d_{in}} = \frac{\omega_{in}}{\omega_{out}} = \frac{C_{out}}{C_{in}}. \tag{446}$$

The flywheel angular speed is written as

$$\omega_{flywheel} = \frac{d_{pulley,motor}\omega_{motor}}{d_{pulley,flywheel}}. \tag{447}$$

The flywheel pulley diameter is written as

$$d_{pulley,flywheel} = \frac{\omega_{generator} d_{pulley,generator}}{\omega_{flywheel}}. \tag{448}$$

The generator pulley diameter is written as

$$d_{pulley,generator} = \frac{\omega_{flywheel} d_{pulley,flywheel}}{\omega_{generator}}. \tag{449}$$

The motor pulley diameter is written as

$$d_{pulley,motor} = \frac{\omega_{flywheel} d_{pulley,flywheel}}{\omega_{motor}}. \tag{450}$$

With

$$\theta_{flywheel}(t) = \frac{\pi}{2}, \tag{451}$$

the flywheel mass is written as

$$M_{flywheel} = \frac{P_{e,generator}}{2\omega_{generator} g r_{flywheel}}. \tag{452}$$

### 9.13 Loss Model

The total mechanical loss torque may be represented as

$$\tau_{\text{loss}} = \tau_{\text{friction}} + \tau_{\text{windage}} + \tau_{\text{bearing}} + \tau_{\text{other}}. \tag{453}$$

A simplified viscous-friction model is

$$\tau_{\text{friction}} = c\omega. \tag{454}$$

A more general model may include Coulomb friction :

$$\tau_{\text{loss}} = \tau_c \,\text{sgn}(\omega) + c\omega + \tau_{\text{windage}}(\omega). \tag{455}$$

For aerodynamic windage, a quadratic approximation may be used :

$$\tau_{\text{windage}} = k_w \omega |\omega|. \tag{456}$$

The parameters of these loss models must ultimately be determined from experimental measurements or justified engineering estimates.

### 9.14 Complete Mechanical Equation

Combining the principal mechanical contributions gives

$$I_{\text{eq}}\dot{\omega} = \tau_{\text{motor}} + \tau_{\text{grav}} - \tau_{\text{gen}} - \tau_{\text{loss}}. \tag{457}$$

The corresponding angular acceleration is

$$\dot{\omega} = \frac{\tau_{\text{motor}} + \tau_{\text{grav}} - \tau_{\text{gen}} - \tau_{\text{loss}}}{I_{\text{eq}}}. \tag{458}$$

This equation constitutes the principal mechanical state equation of the simplified machine model.

### 9.15 Energy Balance

The total mechanical energy may be expressed as

$$E_{\text{mech}} = E_{\text{grav}} + E_{\text{rot}} + E_{\text{fw}}. \tag{459}$$

For a collection of masses,

$$E_{\text{grav}} = \sum_i m_i g h_i. \tag{460}$$

The rotational kinetic energy is

$$E_{\text{rot}} = \frac{1}{2} I_{\text{eq}} \omega^2. \tag{461}$$

The general energy balance is

$$\frac{dE_{\text{mech}}}{dt} = P_{\text{motor}} + P_{\text{grav}} - P_{\text{gen}} - P_{\text{loss}}. \tag{462}$$

For a complete accounting interval,

$$E_{\text{in}} = \Delta E_{\text{stored}} + E_{\text{out}} + E_{\text{loss}}. \tag{463}$$

For a closed cycle, gravitational potential energy must return to its initial value if the mechanical configuration is restored to its initial state. This energy balance is an essential consistency condition for the numerical model.

## 9.16 State-Space Representation

For numerical simulation, the machine can be represented by a state vector

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \\ \mathbf{x}_m \\ \mathbf{x}_g \end{bmatrix}, \tag{464}$$

where $\mathbf{x}_m$ and $\mathbf{x}_g$ denote the internal states of the motor and generator models.
The general state equation is

$$\dot{\mathbf{x}} = \mathbf{f}\left(\mathbf{x}, \mathbf{u}, \mathbf{p}, t\right), \tag{465}$$

where

$$\mathbf{x} : \text{system state vector}, \tag{466}$$
$$\mathbf{u} : \text{control input vector}, \tag{467}$$
$$\mathbf{p} : \text{physical parameter vector}. \tag{468}$$

The output vector may be defined as

$$\mathbf{y} = \mathbf{h}\left(\mathbf{x}, \mathbf{u}, \mathbf{p}, t\right), \tag{469}$$

and may include angular velocity, angular acceleration, torque, voltage, current, electrical power, stored energy, and dissipated energy.

## 9.17 Model Hierarchy

The machine model is intended to be developed progressively :

— an ideal mechanical model containing masses, inertias, geometry, gravity, and applied torques ;
— a dissipative mechanical model including losses and transmission effects ;
— an electromechanical model including motor and generator electrical dynamics ;
— an experimentally informed model including measured parameters, uncertainty, and control-system dynamics.

This hierarchical approach makes it possible to identify the physical origin of each predicted effect before additional complexity is added.

## 9.18 Model Validation Requirement

The mathematical model must be validated independently of any desired performance outcome. Numerical simulations shall not be considered evidence of physical feasibility unless the equations, parameters, initial conditions, and conservation laws have been independently verified.

The following quantities shall be monitored during simulation :

— total mechanical energy ;
— gravitational potential energy ;
— rotational kinetic energy ;
— flywheel energy ;
— motor electrical input power ;
— generator electrical output power ;
— mechanical losses ;
— angular momentum ;
— instantaneous torque balance ;

— cumulative energy-balance residual.

A physically consistent simulation should satisfy the corresponding energy and momentum balances within the declared numerical tolerance of the integration method.

### 9.19 Scope of the Present Model

The equations presented in this chapter define a system-level framework rather than a final validated representation of a physical machine.

The numerical values of the parameters, the exact geometry of the Chas Campbell mechanism, the detailed transmission architecture, and the electrical characteristics of the motor-generator system must be specified separately.

The purpose of the model is to provide a transparent mathematical framework in which each proposed mechanism can be tested against Newtonian mechanics, rotational dynamics, and conservation of energy.

Subsequent chapters will use this framework to define the simulation methodology, calculate system behaviour under specified initial conditions, and evaluate the resulting performance quantitatively.

## 10 Simulations

This chapter defines the numerical simulation methodology used to investigate the mathematical model of the Chas Campbell Gravitational Engine.

The purpose of the simulations is not to assume that the proposed machine produces net energy, but to determine its predicted mechanical and electrical behaviour from explicitly defined equations, parameters, and initial conditions.

### 10.1 Simulation Objectives

The numerical simulations have four principal objectives :

— evaluate the time evolution of the mechanical system ;

— determine the instantaneous gravitational and rotational torques ;

— quantify the transfer and storage of mechanical energy ;

— evaluate the electrical power exchanged by the motor-generator subsystem.

A secondary objective is to identify operating conditions under which the mathematical model becomes unstable, inefficient, or physically inconsistent.

### 10.2 Numerical State Vector

The minimum mechanical state vector is defined as

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{q}(t) \\ \dot{\mathbf{q}}(t) \end{bmatrix}, \tag{470}$$

where $\mathbf{q}$ represents the generalized coordinates of the machine.

For the simplified single-axis model,

$$\mathbf{x}(t) = \begin{bmatrix} \theta(t) \\ \omega(t) \end{bmatrix}. \tag{471}$$

The corresponding first-order system is

$$\dot{\theta} = \omega, \tag{472}$$

and

$$\dot{\omega} = \frac{\tau_{\text{motor}} + \tau_{\text{grav}} - \tau_{\text{gen}} - \tau_{\text{loss}}}{I_{\text{eq}}}. \tag{473}$$

This representation is directly suitable for numerical integration.

## 10.3 Initial Conditions

The simulation requires an explicitly defined initial state.
The initial angular position is

$$\theta(0) = \theta_0, \tag{474}$$

and the initial angular velocity is

$$\omega(0) = \omega_0. \tag{475}$$

If additional mechanical coordinates are introduced, their initial positions and velocities must also be specified :

$$\mathbf{q}(0) = \mathbf{q}_0, \qquad \dot{\mathbf{q}}(0) = \dot{\mathbf{q}}_0. \tag{476}$$

The initial conditions shall be reported with every simulation result because the predicted trajectory may depend strongly on the initial configuration.

## 10.4 Simulation Parameters

The numerical model is controlled by a parameter vector

$$\mathbf{p} = \left[I_{\text{eq}}, m_1, \ldots, m_n, g, c, k_w, K_t, \eta_{\text{motor}}, \eta_{\text{gen}}, \ldots\right]^T. \tag{477}$$

All parameters used in a simulation must be explicitly documented.
Parameters shall preferably be classified into three categories :

1. experimentally measured parameters ;
2. parameters obtained from manufacturer specifications or reliable references ;
3. assumed parameters used for exploratory analysis.

Assumed parameters must not be presented as experimentally validated properties of the physical machine.

## 10.5 Time Integration

The equations of motion are integrated over a finite simulation interval

$$0 \leq t \leq T_{\text{sim}}. \tag{478}$$

The integration time step is denoted by

$$\Delta t. \tag{479}$$

For sufficiently smooth systems, a numerical integration method such as the fourth-order Runge–Kutta method may be used.
For a general state equation

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t), \tag{480}$$

the fourth-order Runge–Kutta method evaluates

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n), \tag{481}$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{x}_n + \frac{\Delta t}{2}\mathbf{k}_1, t_n + \frac{\Delta t}{2}\right), \tag{482}$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{x}_n + \frac{\Delta t}{2}\mathbf{k}_2, t_n + \frac{\Delta t}{2}\right), \tag{483}$$

$$\mathbf{k}_4 = \mathbf{f}\left(\mathbf{x}_n + \Delta t\mathbf{k}_3, t_n + \Delta t\right). \tag{484}$$

The state is then updated according to

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{\Delta t}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right). \tag{485}$$

The integration method and numerical tolerances shall be recorded for reproducibility.

## 10.6 Torque Evaluation

At every simulation time step, the principal torque contributions are calculated independently. The gravitational torque is

$$\tau_{\text{grav}}(t) = \tau_{\text{grav}}\left(\mathbf{q}(t)\right). \tag{486}$$

The motor torque is

$$\tau_{\text{motor}}(t) = \tau_{\text{motor}}\left(\mathbf{x}_m(t), \mathbf{u}(t)\right). \tag{487}$$

The generator reaction torque is

$$\tau_{\text{gen}}(t) = \tau_{\text{gen}}\left(\mathbf{x}_g(t), \omega(t)\right). \tag{488}$$

The total resisting torque is

$$\tau_{\text{res}} = \tau_{\text{gen}} + \tau_{\text{loss}}. \tag{489}$$

The net accelerating torque is consequently

$$\tau_{\text{net}} = \tau_{\text{motor}} + \tau_{\text{grav}} - \tau_{\text{res}}. \tag{490}$$

The angular acceleration follows directly :

$$\alpha(t) = \frac{\tau_{\text{net}}(t)}{I_{\text{eq}}}. \tag{491}$$

## 10.7 Energy Calculation

The simulation shall calculate the principal energy components independently of the equations used to calculate acceleration.

The gravitational potential energy is

$$E_{\text{grav}}(t) = \sum_i m_i g h_i(t). \tag{492}$$

The rotational kinetic energy is

$$E_{\text{rot}}(t) = \frac{1}{2}I_{\text{eq}}\omega(t)^2. \tag{493}$$

For an explicitly modelled flywheel,

$$E_{\text{fw}}(t) = \frac{1}{2} I_{\text{fw}} \omega_{\text{fw}}(t)^2. \tag{494}$$

The total stored mechanical energy is therefore represented by

$$E_{\text{stored}} = E_{\text{grav}} + E_{\text{rot}} + E_{\text{fw}}. \tag{495}$$

## 10.8 Power Calculation

Instantaneous mechanical power associated with a torque is

$$P(t) = \tau(t)\omega(t). \tag{496}$$

Consequently, the gravitational mechanical power is

$$P_{\text{grav}} = \tau_{\text{grav}} \omega. \tag{497}$$

The motor mechanical input is

$$P_{\text{motor,mech}} = \tau_{\text{motor}} \omega, \tag{498}$$

while the mechanical power extracted by the generator is

$$P_{\text{gen,mech}} = \tau_{\text{gen}} \omega. \tag{499}$$

Mechanical losses are represented by

$$P_{\text{loss}} = \tau_{\text{loss}} \omega. \tag{500}$$

Electrical power is calculated independently as

$$P_{\text{elec}} = VI. \tag{501}$$

The distinction between mechanical and electrical power is maintained throughout the simulation in order to avoid incorrectly identifying stored or transferred energy as newly generated energy.

## 10.9 Energy Balance Verification

An essential simulation diagnostic is the energy residual.
Let

$$E_{\text{mech}}(t) \tag{502}$$

denote the total mechanical energy.
The numerical energy balance can be expressed as

$$R_E(t) = E_{\text{mech}}(t) - E_{\text{mech}}(0) - \int_0^t \left(P_{\text{motor}} + P_{\text{grav}} - P_{\text{gen}} - P_{\text{loss}}\right) d\tau. \tag{503}$$

For a numerically consistent implementation,

$$R_E(t) \to 0 \tag{504}$$

as the integration accuracy is increased, subject to modelling approximations and numerical error.
The energy residual shall therefore be monitored throughout each simulation.

## 10.10 Angular Momentum Verification

Angular momentum provides a second independent consistency check.

For the equivalent rotational system,

$$L = I_{\text{eq}}\omega. \tag{505}$$

The applied external torque satisfies

$$\frac{dL}{dt} = \tau_{\text{net}}. \tag{506}$$

The numerical solution shall be checked against this relationship.

Any persistent discrepancy between the calculated torque and the observed rate of change of angular momentum indicates an implementation error, an inconsistent model, or an inadequately resolved numerical integration.

## 10.11 Cycle Analysis

If the machine is intended to operate cyclically, simulation results shall be evaluated over complete mechanical cycles.

Let the cycle period be

$$T_c. \tag{507}$$

The net change in mechanical energy over one complete cycle is

$$\Delta E_{\text{cycle}} = E(t + T_c) - E(t). \tag{508}$$

For a perfectly periodic conservative mechanical configuration,

$$\Delta E_{\text{cycle}} = 0 \tag{509}$$

in the absence of external energy input and losses.

When losses and external electrical input are present, the measured cycle energy must satisfy the corresponding energy balance.

Cycle analysis is therefore more informative than observing a temporary increase in rotational speed during only part of a cycle.

## 10.12 Parameter Sweeps

The influence of uncertain or design parameters may be investigated using parameter sweeps.

For a parameter $p$, a set of simulations may be defined as

$$p \in [p_{\min}, p_{\max}]. \tag{510}$$

For each value of $p$, quantities such as maximum angular velocity, average generator power, total energy loss, and cycle energy balance can be calculated.

This approach allows sensitivity to be quantified without assuming a single parameter value to be exact.

## 10.13 Sensitivity Analysis

The sensitivity of an output quantity $y$ to a parameter $p$ may be approximated numerically by

$$S_p = \frac{\partial y}{\partial p}. \tag{511}$$

A normalized sensitivity coefficient may also be defined as

$$S_p^* = \frac{p}{y}\frac{\partial y}{\partial p}. \tag{512}$$

Sensitivity analysis is particularly important for parameters such as mass distribution, moment of inertia, friction coefficients, geometric dimensions, and motor-generator efficiency.

## 10.14 Numerical Convergence

A simulation result should not be accepted solely because a numerical solver produces a trajectory.

The time step shall be progressively reduced to verify convergence.

For example,

$$\Delta t, \qquad \frac{\Delta t}{2}, \qquad \frac{\Delta t}{4} \tag{513}$$

may be compared.

If the principal quantities of interest converge toward stable values, the numerical solution is considered sufficiently resolved for the chosen model.

The convergence of energy and momentum residuals shall also be checked.

## 10.15 Simulation Scenarios

The research program should evaluate several distinct simulation scenarios.

### 10.15.1 Ideal Mechanical Model

The first scenario excludes losses and electrical conversion inefficiencies.

This provides a baseline for studying the purely mechanical behaviour of the proposed geometry.

### 10.15.2 Mechanical Loss Model

The second scenario introduces bearing friction, viscous damping, windage, and other relevant mechanical losses.

This determines how the idealized dynamics change when realistic resistive effects are introduced.

### 10.15.3 Motor-Generator Model

The third scenario introduces the motor and generator models.

The simulation then evaluates the interaction between mechanical rotation and electrical power conversion.

#### 10.15.4 Full System Model

The final scenario combines gravitational dynamics, rotational dynamics, flywheel storage, motor-generator behaviour, mechanical losses, and control inputs.

This represents the highest-fidelity model available within the research framework.

### 10.16 Reproducibility

Every simulation shall be reproducible from the source code and documented parameters.

The simulation record should include :

— model version ;
— parameter set ;
— initial conditions ;
— integration method ;
— simulation duration ;
— time step or solver tolerances ;
— software version ;
— generated data files ;
— generated figures ;
— numerical validation results.

Simulation outputs should be stored separately from source code.

The research repository therefore distinguishes between model source files, input data, processed data, simulation outputs, and generated figures.

### 10.17 Interpretation of Simulation Results

Simulation results shall be interpreted in accordance with the assumptions and limitations of the mathematical model.

A transient increase in angular velocity does not demonstrate sustained energy production.

Similarly, a positive instantaneous gravitational power,

$$P_{\mathrm{grav}} > 0, \tag{514}$$

does not imply positive net energy production over a complete cycle.

The relevant quantity for assessing the machine as a complete system is the integrated energy balance over the specified operating interval.

Accordingly, the principal performance criterion is defined as

$$E_{\mathrm{net}} = E_{\mathrm{out}} - E_{\mathrm{in}}, \tag{515}$$

with all relevant external energy flows explicitly included.

A complete assessment must distinguish between temporary energy transfer, energy storage, energy conversion, and genuinely external energy input.

### 10.18 Simulation Outputs

The following quantities shall be recorded as functions of time :

— angular position $\theta(t)$ ;
— angular velocity $\omega(t)$ ;
— angular acceleration $\alpha(t)$ ;
— gravitational torque $\tau_{\mathrm{grav}}(t)$ ;
— motor torque $\tau_{\mathrm{motor}}(t)$ ;
— generator torque $\tau_{\mathrm{gen}}(t)$ ;
— mechanical loss torque $\tau_{\mathrm{loss}}(t)$ ;

— gravitational potential energy ;
— rotational kinetic energy ;
— flywheel energy ;
— motor electrical power ;
— generator electrical power ;
— total energy residual.

These quantities provide the basis for the analysis presented in the following chapters.

### 10.19 Limitations

The numerical simulation cannot establish physical validity beyond the validity of the underlying mathematical model.

In particular, simulations cannot compensate for :

— incorrect assumptions ;
— inaccurate geometry ;
— incorrect mass distributions ;
— underestimated mechanical losses ;
— incorrect motor or generator parameters ;
— numerical implementation errors ;
— violations of conservation laws.

Consequently, numerical results shall be regarded as model predictions until they are supported by analytical verification and, where appropriate, experimental measurements.

The simulation framework defined in this chapter provides the computational basis for the quantitative results presented in the next chapter.

## 11 Results

This chapter presents the results obtained from the mathematical and numerical models defined in the preceding chapters.

At the present stage of the research, the purpose of the results is to document the behaviour predicted by the model rather than to claim experimental validation of the Chas Campbell Gravitational Engine.

### 11.1 Overview of the Results

The numerical framework provides time-dependent values for the principal mechanical and electrical quantities of the system.

The quantities of interest include

— angular position ;
— angular velocity ;
— angular acceleration ;
— gravitational torque ;
— motor torque ;
— generator reaction torque ;
— mechanical losses ;
— gravitational potential energy ;
— rotational kinetic energy ;
— flywheel energy ;
— electrical input power ;
— electrical output power.

The results are evaluated using the governing equations established in the mathematical model and the numerical methodology described in the simulation chapter.

## 11.2 Mechanical Response

The principal mechanical response is the time evolution of the angular position,

$$\theta(t), \tag{516}$$

and angular velocity,

$$\omega(t). \tag{517}$$

The angular acceleration is obtained from

$$\alpha(t) = \frac{\tau_{\text{motor}}(t) + \tau_{\text{grav}}(t) - \tau_{\text{gen}}(t) - \tau_{\text{loss}}(t)}{I_{\text{eq}}}. \tag{518}$$

The resulting trajectory provides a direct indication of whether the mechanical assembly accelerates, decelerates, or approaches a quasi-steady operating condition.

A temporary increase in angular velocity must not, by itself, be interpreted as evidence of sustained energy production. The velocity response must be considered together with the complete energy balance.

## 11.3 Gravitational Torque

The gravitational torque varies as a function of the instantaneous configuration of the mechanism.

The simulated gravitational torque is represented by

$$\tau_{\text{grav}}(t) = \tau_{\text{grav}}\left(\mathbf{q}(t)\right). \tag{519}$$

For a cyclic mechanism, the torque may change sign during a complete cycle.

The corresponding mechanical power is

$$P_{\text{grav}}(t) = \tau_{\text{grav}}(t)\omega(t). \tag{520}$$

Consequently, periods of positive gravitational power may be followed by periods of negative gravitational power.

The integrated gravitational contribution over a complete cycle is therefore a more meaningful quantity than its instantaneous maximum.

## 11.4 Rotational Energy

The rotational kinetic energy of the equivalent rotating assembly is

$$E_{\text{rot}}(t) = \frac{1}{2} I_{\text{eq}}\omega(t)^2. \tag{521}$$

An increase in rotational kinetic energy corresponds to acceleration of the rotating system, while a decrease corresponds to deceleration.

The change in rotational energy between two simulation times is

$$\Delta E_{\text{rot}} = E_{\text{rot}}(t_2) - E_{\text{rot}}(t_1). \tag{522}$$

This quantity is compared with the integrated torque and power contributions to verify consistency between the dynamic and energetic representations.

## 11.5 Flywheel Energy

The flywheel stores rotational kinetic energy according to

$$E_{\text{fw}}(t) = \frac{1}{2} I_{\text{fw}} \omega_{\text{fw}}(t)^2. \tag{523}$$

During acceleration, the flywheel may absorb mechanical energy.
During deceleration, it may release part of the previously stored energy.
The net change in flywheel energy over a complete operating cycle is

$$\Delta E_{\text{fw}} = E_{\text{fw}}(t + T_c) - E_{\text{fw}}(t). \tag{524}$$

If the mechanical state returns to its initial condition,

$$\Delta E_{\text{fw}} \approx 0 \tag{525}$$

apart from numerical error and any change intentionally imposed by the operating conditions. The flywheel therefore acts as an energy-storage element rather than an independent energy source.

## 11.6 Motor Contribution

The mechanical contribution of the motor is represented by

$$P_{\text{motor,mech}} = \tau_{\text{motor}} \omega. \tag{526}$$

The corresponding electrical input is

$$P_{\text{motor,elec}} = V_m i_m. \tag{527}$$

The total motor energy supplied during a simulation interval is

$$E_{\text{motor,in}} = \int_{t_0}^{t_1} P_{\text{motor,elec}}(t)\, dt. \tag{528}$$

This quantity must be included when evaluating the net energy balance of the complete machine.

## 11.7 Generator Output

The generator extracts mechanical power according to

$$P_{\text{gen,mech}} = \tau_{\text{gen}} \omega. \tag{529}$$

The electrical output power is

$$P_{\text{gen,elec}} = V_g i_g. \tag{530}$$

The total electrical energy delivered during a simulation interval is

$$E_{\text{gen,out}} = \int_{t_0}^{t_1} P_{\text{gen,elec}}(t)\, dt. \tag{531}$$

The generator output must be compared with all external energy supplied to the system and not solely with the instantaneous gravitational contribution.

## 11.8 Mechanical Losses

Mechanical losses are represented by

$$\tau_{\text{loss}} = \tau_{\text{friction}} + \tau_{\text{windage}} + \tau_{\text{bearing}} + \tau_{\text{other}}. \tag{532}$$

The corresponding power loss is

$$P_{\text{loss}} = \tau_{\text{loss}}\omega. \tag{533}$$

The cumulative mechanical loss over a simulation interval is

$$E_{\text{loss}} = \int_{t_0}^{t_1} P_{\text{loss}}(t)\, dt. \tag{534}$$

These losses are expected to reduce the available mechanical energy and therefore provide an important constraint on the achievable electrical output.

## 11.9 Energy Balance

The principal result used to assess the complete system is the energy balance.

The mechanical energy may be represented by

$$E_{\text{mech}} = E_{\text{grav}} + E_{\text{rot}} + E_{\text{fw}}. \tag{535}$$

The external and internal energy flows are then compared using

$$E_{\text{balance}} = E_{\text{in}} - E_{\text{out}} - \Delta E_{\text{stored}} - E_{\text{loss}}. \tag{536}$$

For a correctly implemented model, the residual should remain close to zero within the expected numerical tolerance :

$$E_{\text{balance}} \approx 0. \tag{537}$$

This provides an independent diagnostic of the numerical implementation.

## 11.10 Cycle Energy Balance

For a cyclic operating regime, the most important quantity is the net energy balance over a complete cycle.

Let

$$T_c \tag{538}$$

denote the cycle period.

The net electrical energy balance may be written as

$$\Delta E_{\text{elec}} = E_{\text{gen,out}} - E_{\text{motor,in}}. \tag{539}$$

However, this quantity alone does not constitute the complete energy balance because changes in gravitational potential energy, stored rotational energy, and losses must also be considered.

The complete cycle balance is therefore

$$E_{\text{in}} + E_{\text{grav,in}} = E_{\text{out}} + E_{\text{loss}} + \Delta E_{\text{stored}}. \tag{540}$$

For a periodic steady state in which the mechanical configuration returns to its initial state,

$$\Delta E_{\text{stored}} = 0. \tag{541}$$

The cycle balance then reduces to

$$E_{\text{in}} + E_{\text{grav,net}} = E_{\text{out}} + E_{\text{loss}}. \tag{542}$$

If the gravitational system also returns to its initial configuration, the net gravitational energy contribution over the cycle is zero :

$$E_{\text{grav,net}} = 0. \tag{543}$$

The resulting balance becomes

$$E_{\text{in}} = E_{\text{out}} + E_{\text{loss}}. \tag{544}$$

This relation provides a fundamental consistency criterion for any claimed cyclic operating condition.

### 11.11 Numerical Energy Residual

The numerical energy residual is defined as

$$R_E(t) = E_{\text{mech}}(t) - E_{\text{mech}}(0) - \int_0^t [P_{\text{motor}} + P_{\text{grav}} - P_{\text{gen}} - P_{\text{loss}}]\, d\tau. \tag{545}$$

For a sufficiently accurate numerical integration,

$$R_E(t) \to 0. \tag{546}$$

The residual should be plotted as a function of time.

A persistent or growing residual may indicate one of the following :

— an error in the equations ;
— an error in the implementation ;
— an incorrect sign convention ;
— an inconsistent parameter definition ;
— insufficient numerical resolution.

The residual is therefore treated as a validation metric rather than as a performance parameter.

### 11.12 Angular Momentum Results

The angular momentum of the equivalent rotating system is

$$L(t) = I_{\text{eq}}\omega(t). \tag{547}$$

The numerical derivative is compared with the applied net torque :

$$\frac{dL}{dt} = \tau_{\text{net}}. \tag{548}$$

Agreement between these quantities provides an independent validation of the mechanical integration.

The comparison is particularly important when the gravitational torque varies significantly during a cycle.

## 11.13 Parameter Sensitivity

The sensitivity analysis described previously can be used to determine which parameters have the greatest influence on the predicted performance.

For a generic output quantity $y$, the normalized sensitivity is

$$S_p^* = \frac{p}{y}\frac{\partial y}{\partial p}. \tag{549}$$

Parameters with large absolute values of $S_p^*$ have a stronger influence on the selected output.

Important candidate parameters include

— mass distribution ;
— moment of inertia ;
— geometric dimensions ;
— friction coefficients ;
— windage coefficients ;
— motor efficiency ;
— generator efficiency ;
— initial angular velocity.

Sensitivity results should be used to identify which physical measurements would provide the greatest improvement in model accuracy.

## 11.14 Numerical Convergence Results

The numerical solution must be checked for convergence with respect to the integration step.

Let the results obtained with three integration steps be

$$\Delta t, \qquad \frac{\Delta t}{2}, \qquad \frac{\Delta t}{4}. \tag{550}$$

A representative numerical quantity $y$ is considered converged when

$$\left| y_{\Delta t/2} - y_{\Delta t/4} \right| \ll \left| y_{\Delta t} \right|. \tag{551}$$

The precise convergence criterion shall be defined according to the accuracy requirements of the corresponding simulation.

## 11.15 Interpretation of the Present Results

The present numerical framework demonstrates how the proposed machine can be analysed using standard mechanical and electromechanical principles.

The results must nevertheless be interpreted within the limits of the model.

In particular, the following observations are not sufficient by themselves to demonstrate net energy production :

— an increase in angular velocity ;
— a positive instantaneous gravitational torque ;
— a positive instantaneous gravitational power ;
— a temporary increase in flywheel energy ;
— a positive instantaneous generator output.

Each of these quantities represents a local or transient property of the system.

The decisive criterion for a cyclic machine is the complete energy balance over the relevant operating interval.

### 11.16 Current Status of the Results

At the current stage of the research, the numerical results should be regarded as model-dependent predictions.

No numerical result shall be interpreted as experimental confirmation unless the corresponding parameter values and measurements have been independently established.

The simulation framework provides a basis for progressively replacing assumed parameters with measured values.

This distinction between theoretical prediction, numerical simulation, and experimental observation is maintained throughout the research.

### 11.17 Required Result Figures

The following figures are recommended for the completed simulation campaign :

1. angular position as a function of time ;
2. angular velocity as a function of time ;
3. angular acceleration as a function of time ;
4. gravitational torque as a function of angular position ;
5. individual torque contributions as a function of time ;
6. gravitational potential energy as a function of time ;
7. rotational kinetic energy as a function of time ;
8. flywheel energy as a function of time ;
9. motor input power as a function of time ;
10. generator output power as a function of time ;
11. cumulative input and output energy ;
12. total energy residual ;
13. angular momentum verification ;
14. parameter sensitivity ;
15. numerical convergence.

These figures should be generated directly from the simulation data rather than manually constructed.

### 11.18 Summary

The results framework establishes a quantitative basis for evaluating the Chas Campbell Gravitational Engine.

The central requirement is that mechanical motion, gravitational interaction, energy storage, electrical conversion, and losses be considered simultaneously.

The results therefore do not rely on a single performance indicator. Instead, the physical interpretation is based on the simultaneous verification of torque balance, energy balance, angular momentum, numerical convergence, and parameter sensitivity.

The discussion of the physical meaning and limitations of these results is presented in the following chapter.

## 12 Discussion

This chapter discusses the theoretical implications of the mathematical model developed in the preceding chapters. The objective is to distinguish clearly between the internal dynamics of the proposed mechanism and the question of whether the complete system can provide a net positive energy output.

## 12.1 Interpretation of the Model

The model developed in this research describes a coupled mechanical system involving gravitational forces, rotational motion, a flywheel, and a motor-generator subsystem. The equations introduced in the preceding chapters are intended to provide a consistent mathematical representation of the proposed architecture.

The presence of gravitational forces does not, by itself, imply the continuous production of net mechanical energy. Gravitational potential energy is a state-dependent quantity, and any energy extracted during a downward displacement must be considered together with the energy required to restore the corresponding mass distribution to its initial state.

Consequently, the complete cycle must be analyzed rather than isolated portions of the motion.

## 12.2 Energy Conservation

For an idealized closed mechanical system, the total energy satisfies the conservation relation

$$E_{\text{total}} = E_{\text{kinetic}} + E_{\text{potential}} + E_{\text{electrical}} + E_{\text{other}}, \tag{552}$$

with appropriate accounting for energy transfers between the different subsystems.

For the gravitational component, the potential energy of a mass $m$ at height $h$ relative to a chosen reference level is

$$E_g = mgh. \tag{553}$$

A decrease in gravitational potential energy can therefore produce mechanical work. However, if the mechanism subsequently returns the mass to its original configuration, the energy required for this restoration must also be included in the cycle balance.

The relevant quantity for evaluating the proposed engine is therefore not the instantaneous mechanical power but the net energy balance over a complete operating cycle.

## 12.3 Flywheel Dynamics

The flywheel provides an energy-storage mechanism through its rotational kinetic energy,

$$E_{\text{flywheel}} = \frac{1}{2} I\omega^2, \tag{554}$$

where $I$ is the moment of inertia and $\omega$ is the angular velocity.

The corresponding mechanical power associated with a changing angular velocity is

$$P_{\text{rot}} = \tau\omega, \tag{555}$$

where $\tau$ denotes the applied torque.

The flywheel may therefore smooth transient variations in torque and angular velocity. It may also permit energy to be temporarily stored and released at different points in the operating cycle.

This storage function should not, however, be interpreted as an independent source of energy. The energy stored in the flywheel must ultimately originate from an energy transfer elsewhere in the system.

### 12.4 Motor-Generator Interaction

The motor-generator subsystem introduces an additional energy-conversion path. In an idealized representation, mechanical and electrical powers can be related through

$$P_{\text{mech}} = \tau\omega \tag{556}$$

and

$$P_{\text{elec}} = VI, \tag{557}$$

where $V$ and $I$ are the electrical voltage and current.

For a real device, conversion losses must be included. A simplified efficiency representation is

$$P_{\text{out}} = \eta P_{\text{in}}, \qquad 0 < \eta < 1. \tag{558}$$

If the same motor-generator subsystem is used repeatedly to transfer energy between mechanical and electrical domains, the cumulative effect of these efficiencies can become significant.

For example, if an energy quantity is converted from mechanical to electrical form and subsequently converted back to mechanical form, the resulting energy is approximately

$$E_{\text{return}} = \eta_{\text{gen}}\eta_{\text{motor}}E_{\text{initial}}, \tag{559}$$

before additional mechanical losses are considered.

### 12.5 Gravitational Energy Transfer

The gravitational subsystem remains central to the proposed architecture. The instantaneous gravitational force acting on a mass near the Earth's surface is approximated by

$$F_g = mg. \tag{560}$$

For a displacement $dh$, the corresponding differential work is

$$dW_g = -mg\,dh, \tag{561}$$

subject to the selected sign convention.

The total gravitational work between two positions is therefore

$$W_g = -mg(h_2 - h_1). \tag{562}$$

This relation provides a direct means of calculating the mechanical energy associated with a prescribed gravitational displacement.

The important question for the proposed engine is whether the complete kinematic cycle produces an energy asymmetry that remains after all restoration operations, internal energy transfers, and losses have been included.

### 12.6 Cycle-Level Analysis

A meaningful evaluation of the system requires the definition of a complete cycle. Let the initial state of the system be represented by

$$\mathcal{S}_0 = (h_0,\, \theta_0,\, \omega_0,\, x_0,\, \ldots)\,. \tag{563}$$

After one complete cycle, the final state is

$$\mathcal{S}_1 = (h_1,\, \theta_1,\, \omega_1,\, x_1,\, \ldots)\,. \tag{564}$$

For a periodic operating regime, the relevant condition is

$$\mathcal{S}_1 \approx \mathcal{S}_0. \tag{565}$$

The net useful energy delivered during the cycle can then be expressed schematically as

$$E_{\text{net}} = E_{\text{out}} - E_{\text{restoration}} - E_{\text{loss}}. \tag{566}$$

A positive value of $E_{\text{net}}$ would require an identifiable external energy contribution or a previously omitted energy reservoir. For an isolated conservative system, a sustained positive net energy balance would require examination of the model for an inconsistency, missing interaction, or violation of the assumed conservation laws.

### 12.7 Role of Numerical Simulation

Numerical simulation is useful for determining the temporal evolution of the coupled system. It can provide quantities such as

$$h(t), \qquad \theta(t), \qquad \omega(t), \qquad \tau(t), \qquad P(t), \tag{567}$$

and can therefore reveal transient behavior, equilibrium points, oscillations, instabilities, and energy transfers between subsystems.

However, a numerical simulation does not by itself establish the physical validity of the underlying model. The equations implemented in the simulation must first correspond to physically justified assumptions and boundary conditions.

In particular, numerical integration errors, inappropriate constraints, or incomplete accounting of reaction forces can produce apparent energy imbalances. Energy conservation should therefore be monitored explicitly during simulations.

### 12.8 Losses and Non-Ideal Effects

The idealized equations presented in the preceding chapters do not necessarily capture all losses present in a physical implementation. Potential sources of energy dissipation include

— bearing friction,
— aerodynamic drag,
— structural deformation,
— electrical resistance,
— magnetic losses,
— motor and generator inefficiency,
— mechanical transmission losses,
— control-system consumption,
— vibration and acoustic losses.

A more realistic model should introduce these effects explicitly whenever experimental or manufacturer data are available.

A generalized mechanical equation can be written as

$$I\dot{\omega} = \tau_{\text{drive}} - \tau_{\text{load}} - \tau_{\text{loss}}, \tag{568}$$

where $\tau_{\text{loss}}$ represents the aggregate resisting torque.

Similarly, the electrical subsystem can be represented by an effective efficiency or by a more detailed electromechanical model.

### 12.9 Sensitivity to Parameters

The behavior of the proposed system may depend strongly on parameters such as mass, radius, moment of inertia, angular velocity, gravitational acceleration, transmission ratio, and conversion efficiency.

For example, the flywheel energy scales according to

$$E_{\text{flywheel}} \propto I\omega^2. \tag{569}$$

Consequently, changes in angular velocity can have a significant effect on the amount of stored rotational energy.

Likewise, gravitational potential energy scales linearly with both mass and height :

$$E_g \propto mh. \tag{570}$$

Parameter sweeps and sensitivity analyses should therefore be performed before conclusions are drawn from a particular numerical configuration.

### 12.10 Theoretical Limitations

The present research is theoretical and should not be interpreted as an experimental demonstration of a self-sustaining or over-unity energy source.

The mathematical model establishes a framework for analyzing the proposed mechanism. It does not, by itself, demonstrate that the complete machine can generate more energy than is supplied to it.

A rigorous evaluation requires that every external input, stored-energy term, gravitational interaction, mechanical constraint, and dissipative process be explicitly identified.

Particular attention should also be given to reaction forces associated with the mechanical supports and constraints. These forces may perform work on different components even when their contribution is not immediately apparent in a simplified free-body representation.

### 12.11 Requirements for Experimental Validation

A future experimental implementation should independently measure the principal energy flows in the system.

Relevant measurements include

— input electrical power,
— output electrical power,
— shaft torque,
— angular velocity,
— gravitational displacement,
— mass position,
— acceleration,
— operating temperature,
— duration of the complete cycle.

The instantaneous mechanical power should be determined from

$$P_{\text{mech}}(t) = \tau(t)\omega(t). \tag{571}$$

The corresponding energy over a measurement interval is

$$E = \int_{t_0}^{t_1} P(t)\, dt. \tag{572}$$

Independent measurements of input and output energy are essential for determining whether an observed output is genuinely attributable to the proposed mechanism rather than to stored energy, measurement uncertainty, or an unaccounted external input.

### 12.12 Overall Assessment

The mathematical framework developed in this research provides a basis for systematically examining the Chas Campbell Gravitational Engine as a coupled gravitational, rotational, and electromechanical system.

The principal theoretical requirement is a complete energy balance over a closed cycle. Gravitational potential energy, rotational kinetic energy, electrical energy, and dissipative losses must all be included within the same accounting framework.

The model therefore serves primarily as a testable theoretical framework. Its purpose is to determine whether the proposed architecture contains a physically realizable mechanism capable of producing a useful energy output, and to identify precisely where energy enters, is stored, transferred, converted, and dissipated.

Any claim of sustained net energy production must ultimately be supported by a closed-cycle energy balance and, subsequently, by reproducible experimental measurements.

## 13 Conclusion

This research has established a theoretical framework for the analysis of the Chas Campbell Gravitational Engine as a coupled mechanical, gravitational, rotational, and electromechanical system.

The objective of the present work is not to assume a particular outcome, but to formulate the physical and mathematical conditions required to evaluate the proposed mechanism rigorously. The preceding chapters have therefore introduced the principal assumptions, governing equations, gravitational interactions, rotational dynamics, flywheel behavior, motor-generator coupling, machine-level representation, and the framework required for numerical investigation.

A central result of the theoretical framework is that the performance of the complete system must be evaluated over a full operating cycle. The energy associated with gravitational displacement can be expressed as

$$E_g = mgh, \tag{573}$$

while the rotational energy stored in a flywheel is

$$E_{\text{rot}} = \frac{1}{2}I\omega^2. \tag{574}$$

Electrical and mechanical conversion processes must likewise be included in the global energy balance.

For a complete cycle, the relevant quantity is the net energy balance,

$$E_{\text{net}} = E_{\text{out}} - E_{\text{in}}, \tag{575}$$

with all stored-energy variations, restoration work, and losses accounted for consistently.

If the system returns to its initial physical state after one complete cycle, then the changes in internal stored energy must also be included in the balance. In an ideal closed conservative system, the conservation of energy imposes

$$\Delta E_{\text{total}} = 0. \tag{576}$$

This condition provides a fundamental criterion against which the proposed mechanism can be tested.

The theoretical model therefore does not establish, by itself, that the Chas Campbell Gravitational Engine can provide sustained net energy production. Instead, it establishes a framework through which such a hypothesis can be quantitatively tested. Any positive net energy result must be traceable to a clearly identified physical energy source and must remain present after all relevant energy transfers and losses have been included.

Numerical simulation represents the next important stage of the research. The equations developed in this document can be implemented computationally to investigate the temporal behavior of the system under controlled parameter sets. Quantities such as position, velocity, angular velocity, torque, mechanical power, electrical power, and total system energy can be tracked throughout complete operating cycles.

The numerical model should also monitor the conservation of energy explicitly. A representative diagnostic quantity is

$$\epsilon_E(t) = E_{\text{total}}(t) - E_{\text{total}}(0), \tag{577}$$

which should remain consistent with the expected numerical and physical error bounds for a closed idealized system.

Future research should progressively increase the physical fidelity of the model. This includes incorporating experimentally determined friction, bearing losses, aerodynamic drag, electrical resistance, motor-generator efficiency, structural losses, transmission losses, and measurement uncertainties.

Experimental validation will ultimately be required to determine whether the theoretical behavior predicted by the model corresponds to the behavior of a physical implementation. Such validation should independently measure both input and output energy over complete cycles and should account for all external connections to the system.

The principal scientific contribution of the present work is therefore the establishment of a structured and falsifiable methodology for studying the proposed gravitational engine. The architecture can be analyzed using standard principles of mechanics, dynamics, electromechanical conversion, and energy conservation without requiring the desired conclusion to be assumed in advance.

The continuation of this research should consequently focus on three principal objectives :

1. completing and validating the coupled mathematical model ;
2. performing numerical simulations and complete-cycle energy accounting ;
3. comparing theoretical predictions with reproducible experimental measurements.

Only after these stages have been completed can a scientifically justified conclusion be reached regarding the practical energy performance of the Chas Campbell Gravitational Engine.